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SEMICONDUCTOR STRUCTURE AND METHOD OF MANUFACTURING THE SAME

Abstract

Various embodiments of the present disclosure are directed to a semiconductor structure in which circuit components of the semiconductor structure are split amongst various IC dies, which are configured to receive different voltages to reduce noise and/or enhanced performance. As an example, a first integrated circuit (IC) die includes a first semiconductor substrate, and a second IC die includes a second semiconductor substrate. A pixel includes a sensor in the first IC die and a pixel circuit in the second IC die. The pixel circuit includes a plurality of transistors, the sensor includes a first terminal and a second terminal that are both in the first semiconductor substrate, the first terminal is electrically coupled to an input of the pixel circuit, and the second terminal is electrically isolated from individual transistor bodies of the plurality of transistors.

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Background/Summary

REFERENCE TO RELATED APPLICATION [0001] This Application claims the benefit of U.S. Provisional Application No. 63/662,427, filed on Jun. 21, 2024, and is a Continuation-in-Part of U.S. application Ser. No. 17/887,634, filed on Aug. 15, 2022, which claims the benefit of U.S. Provisional Application No. 63/342,659, filed on May 17, 2022, the contents of each of these Applications are hereby incorporated by reference in their entirety.

BACKGROUND
[0002] The semiconductor manufacturing industry continuously seeks to increase functional density. Traditionally, functional density has been increased by scaling down circuit components. However, it is becoming increasingly difficult to continue scaling down circuit components. Therefore, advanced packaging techniques and chip stacking are increasingly employed to increase functional density. Further, as circuit components are scaled down, supply voltage is generally reduced. This reduces power consumption but may be incompatible with or otherwise degrade performance for certain types of circuit components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. The figures are drawn to clearly illustrate relevant aspects of the embodiments. The figures may illustrate relationships between various structures and/or elements within the embodiments. It is noted that the figures are not necessarily drawn to scale. In some instances, the dimensions of various features may be arbitrarily increased or reduced for clarity of discussion.
[0004] FIG. **1** illustrates a circuit diagram of some embodiments of an image sensor that spans three integrated circuit (IC) dies, according to aspects of the present disclosure.
[0005] FIGS. **2**A-**2**C illustrate various circuit diagrams of some alternative embodiments of the image sensor of FIG. **1**.
[0006] FIGS. **3**A and **3**B illustrate various circuit diagrams of some embodiments of the image sensors of FIGS. **1** and **2**A-**2**C in which a current source and an analog-to-digital converter (ADC) further comprise individual transistors.
[0007] FIG. **4** illustrates a cross-sectional view of some embodiments of the image sensor of FIG. **1**.
[0008] FIG. **5** illustrates a plan view of some embodiments of the image sensor of FIG. **4**.

[0009] FIG. **6** illustrates a circuit diagram of some embodiments of an image sensor that spans two IC dies, according to aspects of the present disclosure.
[0010] FIGS. **7**A-**7**C illustrate various circuit diagrams of some alternative embodiments of the image sensor of FIG. **6**.
[0011] FIGS. **8**A and **8**B illustrate various circuit diagrams of some embodiments of the image sensors of FIGS. **6** and **7**A-**7**C in which a current source and an ADC further comprise individual transistors.
[0012] FIG. **9** illustrates a cross-sectional view of some embodiments of the image sensor of FIG. **6**.
[0013] FIG. **10** illustrates a plan view of some embodiments of the image sensor of FIG. **9**.
[0014] FIGS. **11**A and **11**B illustrate various diagrams of some embodiments of an image sensor comprising a plurality of pixels, according to aspects of the present disclosure.
[0015] FIG. **12** illustrates a circuit diagram of some embodiments of a biosensor that spans three IC dies, according to aspects of the present disclosure.
[0016] FIGS. **13**A-**13**G illustrate various circuit diagrams of some alternative embodiments of the biosensor of FIG. **12**.
[0017] FIG. **14** illustrates a circuit diagram of some embodiments of the biosensor of FIG. **12** in which a controller and an ADC further comprise individual transistors.
[0018] FIG. **15** illustrates a cross-sectional view of some embodiments of the biosensor of FIG. **12**.
[0019] FIGS. **16**A and **16**B illustrate cross-sectional views of some embodiments of a first IC die, a second IC die, and third IC die in the biosensor of FIG. **15**.
[0020] FIG. **17** illustrates a plan view of some embodiments of the first IC die of FIGS. **16**A and **16**B.
[0021] FIG. **18** illustrates a cross-sectional view of some alternative embodiments of the biosensor of FIG. **15**.
[0022] FIG. **19** illustrates a cross-sectional view of some embodiments of a first IC die, a second IC die, and a third IC die in the biosensor of FIG. **18**.
[0023] FIG. **20** illustrates a cross-sectional view of some alternative embodiments of the biosensor of FIG. **15**.
[0024] FIGS. **21**A and **21**B illustrate cross-sectional views of some embodiments of a first IC die, a second IC die, and third IC die in the biosensor of FIG. **20**.
[0025] FIG. **22** illustrates a circuit diagram of some embodiments of a biosensor that spans two IC dies, according to aspects of the present disclosure
[0026] FIGS. **23**A-**23**E illustrate various circuit diagrams of some alternative embodiments of the biosensor of FIG. **22**.
[0027] FIG. **24** illustrates a circuit diagram of some embodiments of the biosensor of FIG. **22** in which a controller and an ADC further comprise individual transistors.
[0028] FIG. **25** illustrates a cross-sectional view of some embodiments of the biosensor of FIG. **22**.
[0029] FIG. **26** illustrates a cross-sectional view of some embodiments of a first IC die and a second IC die in the biosensor of FIG. **25**.
[0030] FIGS. **27**A and **27**B illustrate various diagrams of some embodiments of a biosensor comprising a plurality of pixels, according to aspects of the present disclosure.
[0031] FIG. **28** illustrates a block diagram of some embodiments of a method of use according to aspects of the present disclosure

DETAILED DESCRIPTION

[0032] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0033] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In some embodiments, the terms "approximately" and/or "about" can be interpreted as meaning +/−10% or +/−5%, while in other embodiments, the terms "approximately" and/or "about" can be interpreted as meaning within the normal fabrication tolerances of a given fab manufacturing flow.

[0034] A semiconductor structure may include various different types of circuit components formed in one or more integrated circuit (IC) dies. Such circuit components may, for example, include memory cells, photodetectors, transistors, etc. Further, the various different types of circuit components may depend on different supply voltages. Therefore, as the semiconductor structure is scaled down and supply voltage is reduced, challenges may arise.

[0035] Take a complementary metal-oxide semiconductor (CMOS) image sensor (CIS) as an example. The CIS may comprise a first IC die and a second IC die that are stacked and bonded together. The first IC die accommodates an array of pixels, each comprising a photodetector and a pixel circuit, whereas the second IC die accommodates an application-specific integrated circuit (ASIC) common to and electrically coupled individually to the pixels. As the CIS is scaled down, supply voltage may be reduced. However, this may prevent the full transfer of charge from photodetectors to corresponding floating diffusion nodes (FDNs), which may negatively impact quantum efficiency (QE) of the CIS.

[0036] To counter the decrease in supply voltage from the scaling down, a semiconductor structure may employ different reference voltages at different IC dies. For example, continuing with the CIS example, the first IC die may have a negative reference voltage, whereas the second IC die may have a reference voltage that is ground. The negative reference voltage effectively increases the supply voltage at the photodetector to improve charge transfer. However, using different reference voltages at different IC dies may cause noise across the semiconductor structure due to non-tracking between the two reference voltages.

[0037] Various embodiments of the present disclosure are directed towards a semiconductor structure in which circuit components of the semiconductor structure are split amongst various IC dies, which are configured to receive different supply and/or reference voltages. Further, the circuit components are arranged within the various IC dies so minimum supply voltages to the circuit components are met and so noise is reduced.

[0038] As an example, a CIS may comprise photodetectors, corresponding pixel circuits, and an ASIC. The photodetectors may be in a first IC die. The pixel circuits may be split amongst the first IC die and a second IC die with transfer transistors of the pixel circuits being in the first IC die and remaining pixel transistors of the pixel circuits being in the second IC die. The ASIC may be in the second IC die or, alternatively, may be in a third IC die separated from the first IC die by the second IC die. Further, the first IC die may have a negative reference voltage, whereas the second IC die and, where present, the third IC die may have a reference voltage that is ground. The negative reference voltage effectively increases the supply voltage at the photodetectors to allow full charge transfer from the photodetectors.

[0039] The portions of the pixel circuits in the second IC die are responsible for generating the signals that are output from the pixel circuits to the ASIC. Because these portions have the same reference voltage as the ASIC, there is no non-tracking reference noise passing between the pixel circuits and the ASIC. Hence, QE of the CIS is improved.
[0040] With reference to FIG. **1**, a circuit diagram **100** of some embodiments of an image sensor spanning three IC dies, according to aspects of the present disclosure, is provided. The image sensor may, for example, be a CIS or some other suitable type of image sensor. The image sensor includes a first IC die **1**, a second IC die **2**, and a third IC die **3** that are stacked with the second IC die **2** being between and bonded to the first and third IC dies **1**, **3**. Further, the second IC die **2** is electrically coupled to the first and third IC dies **1**, **3** respectively by a first interconnect structure **41** and a second interconnect structure **42**.
[0041] A pixel PX spans the first and second IC dies **1**, **2** and comprises a pinned photodiode **10** and a pixel circuit **20**. In alternative embodiments, the pinned photodiode **10** is some other suitable type of photodetector. The pinned photodiode **10** is in the first IC die **1**, and the pixel circuit **20** is in both the first IC die **1** and the second IC die **2**.
[0042] The pixel circuit **20** is configured to facilitate reset and readout of the pinned photodiode **10**. Further, the pixel circuit **20** comprises a transfer transistor **11** in the first IC die **1** and a plurality of additional pixel transistors (e.g., transistors **21**, **22**, and **23**) in the second IC die **1**. The transfer transistor **11** forms a first pixel-circuit portion **20***f*, whereas the plurality of additional pixel transistors form a second pixel-circuit portion **20***s*.
[0043] The transfer transistor **11** is electrically coupled from a cathode C.sub.10 of the pinned photodiode **10** to a FDN. Further, an anode A.sub.10 of the pinned photodiode **10** and a body B.sub.11 of the transfer transistor **11** are electrically coupled together and configured to receive a first reference voltage V**1**. In some embodiments, reference voltage as used herein may also be referred to as ground level, ground point, or the like. The transfer transistor **11** is configured to selectively transfer charge that accumulates at the pinned photodiode **10** to the FDN.
[0044] The second pixel-circuit portion **20***s* is electrically coupled to the FDN (more generally, the first pixel-circuit portion **20***f*) via the first interconnect structure **41**. Further, individual transistor bodies (e.g., body B.sub.21, body B.sub.22, and body B.sub.23) of the second pixel-circuit portion **20***s* are electrically coupled together (e.g., via a substrate of the second IC die **2**) and are configured to receive a second reference voltage V**2**. The second pixel-circuit portion **20***s* is configured to reset the FDN and the pinned photodiode **10** and to generate an output signal based on the amount of charge transferred to the FDN.
[0045] An ASIC **30** is in the third IC die **3** and electrically coupled to the pixel circuit **20** by the second interconnect structure **42**. The ASIC **30** may, for example, provide control over operation of the pixel PX and/or may, for example, perform analog-to-digital conversion (ADC), image signal processing, and the like on an output of the pixel circuit **20**. The ASIC **30** comprises an ADC **31** and a current source **32**. The ADC **31** and the current source **32** (more generally the ASIC **30**) are configured to receive the second reference voltage V**2**. For example, the ADC **31** and the current source **32** may comprise individual transistors with individual bodies configured to receive the second reference voltage V**2**.
[0046] In some embodiments, the pinned photodiode **10** may be generalized as a sensor or the like. In some embodiments, the pixel circuit **20** may be generalized as an interface circuit for the sensor. In some embodiments, the ASIC **30** may be generalized as a control and/or processing circuit for the interface circuit and/or the sensor.
[0047] Performance of the ADC **31** may be characterized by its signal-to-noise ratio (SNR). Further, it has been appreciated that different reference voltages (e.g., ground levels) respectively at the second pixel-circuit portion **20***s* and the ASIC **30** may significantly degrade the SNR. For example, supposing the second pixel-circuit portion **20***s* had the first reference voltage V**1** and the ASIC **30** had the second reference voltage V**2**, noise may arise and degrade the SNR.

[0048] A reference voltage (more generally any voltage) is subject to (random) variations. Such variations may, for example, correspond to stray voltages and/or may, for example, arise from the resistance of bond wires, conductive traces, etc. in the IC dies. With different reference voltages, the variations in one reference voltage are different than, and hence don't track, the variations in the one or more other reference voltages. This non-tracking leads to noise that negatively affects the performance (e.g., the SNR) of the ADC **31**.
[0049] Because the second pixel-circuit portion **20***s* and the ADC **31** are configured to receive the second reference voltage V**2** (e.g., the same reference voltage), these circuit components have the same reference voltage. Hence, the ADC **31** has the same reference-voltage variations as the second pixel-circuit portion **20***s*. This prevents noise induced by different reference voltages respectively at the second pixel-circuit portion **20***s* and the ASIC **30** and enhances the performance of the ADC **31**.
[0050] In some embodiments, the first reference voltage V**1** is negative, whereas the second reference voltage V**2** is ground or otherwise greater than the first reference voltage V**1**. For example, the first reference voltage V**1** may be −1.3 volts (V), whereas the second reference voltage V**2** may be 0V. Other suitable voltages are, however, amenable. As a result, the pinning voltage of the pinned photodiode **10** and the turn-on voltage of the transfer transistor **11** are relative to the first reference voltage V**1** rather than the second reference voltage V**2**. This effectively increases these voltages (e.g., pinning voltage et al.) by a magnitude of the potential difference between the first and second reference voltages V**1**, V**2**.
[0051] The affective increase in voltages at the pinned photodiode **10** and the transfer transistor **11** allows the supply voltage to be reduced and/or performance of the image sensor to be enhanced. For example, the reduced supply voltage reduces power consumption and may, for example, coincide with scaling down of the image sensor. As another example, the increased pinning voltage increases the full well capacity (FWC) of the pinned photodiode **10**, whereas the increased turn-on voltage allows full transfer of charge from the pinned photodiode **10** to the FDN, to collectively enhance the performance of the image sensor.
[0052] With continued reference to FIG. **1**, the pinned photodiode **10** is configured to convert radiation that enters the image sensor to an electrical signal. When incident light (containing photons of sufficient energy) strikes the pinned photodiode **10**, an electron-hole pair is created. The pinned photodiode **10** includes the cathode C.sub.10 and the anode A.sub.10. Further, the pinned photodiode **10** is connected to the transfer transistor **11**.
[0053] The transfer transistor **11** includes a gate G.sub.11, a source S.sub.11, a drain D.sub.11, and the body B.sub.11. In some embodiments, the transfer transistor **11** may be an n-channel metal-oxide-semiconductor (NMOS) transistor. The source S.sub.11 is connected to the cathode is C.sub.10 of the pinned photodiode **10**, and the body B.sub.11 is connected to the anode A.sub.10 of the pinned photodiode **10**. As noted above, the anode A.sub.10 and the body B.sub.11 are configured to receive the first reference voltage V**1**. For example, the first reference voltage V**1** may be applied to the anode A.sub.10 and the body B.sub.11 to bias the anode A.sub.10 and the body B.sub.11 with the first reference voltage V**1**. The first reference voltage V**1** may, for example, be negative.
[0054] The transfer transistor **11** is turned ON when a high voltage is applied onto the gate G.sub.11, while the transfer transistor **11** is turned OFF when a low voltage is applied onto the gate G.sub.11. The high and low voltages may, for example, correspond to voltages respectively above and below a threshold voltage of the transfer transistor **11**. By biasing the anode A.sub.10 and the body B.sub.11 with a negative voltage, the ground level of the transfer transistor **11** is more negative than when biased at 0V (ground). Thus, the voltage applied to the gate G.sub.11 may be lower to comply with the lower supply voltages, which reduce power consumption.
[0055] The drain D.sub.11 of the transfer transistor **11** is a FDN of the pixel PX. The FDN is configured to store charges (or carriers) transferred from the pinned photodiode **10** through a channel of the transfer transistor **11**. The potential of the FDN is monitored by a source-follower

transistor **22** of second pixel-circuit portion **20***s*.

[0056] The pinned photodiode **10** is included in a substrate of the first IC die **1**. Further, the transfer transistor **11** is included on the substrate. The substrate is configured to receive the first reference voltage V**1**. For example, the first reference voltage V**1** may be applied to a bulk region of the substrate to bias the first substrate. In some embodiments, an array of pinned photodiodes, including the pinned photodiode **10**, is in the substrate. Further, in some embodiments, an array of transfer transistors, including the transfer transistor **11**, is on the first substrate and each transfer transistor controls one of the pinned photodiodes.

[0057] In some embodiments, the pinned photodiode **10** may be formed by a heavily-doped capping layer with a first dopant type, a lightly-doped buried layer with a second dopant type, and a lightly doped bottom layer with the first dopant type. In some embodiments, the first dopant type is a p-type and the second dopant type is n-type. The pinned photodiode **10** may have a p+np vertical structure in the substrate of the first IC die **1**.

[0058] The second pixel-circuit portion **20***s* comprises a reset transistor **21**, the source-follower transistor **22**, and a row-select transistor **23**. Alternatively, there may be more or less transistors. The reset transistor **21** includes a gate G.sub.21, a source S.sub.21, a drain D.sub.21, and a body B.sub.21 and is connected to the transfer transistor **11** and the source-follower transistor **22**. The source-follower transistor **22** includes a gate G.sub.22, a source S.sub.22, a drain D.sub.22, and a body B.sub.22 and is connected to the row-select transistor **23**. The row-select transistor **23** includes a gate G.sub.23, a source S.sub.23, a drain D.sub.23, and a body B.sub.23 and is connected to the ASIC **30**.

[0059] The gate G.sub.21 of the reset transistor **21** is configured to receive a control signal (e.g., from the ASIC **30**) to control the state of the reset transistor **21**. When ON, the reset transistor **21** resets the FDN by pulling it to a supply voltage V.sub.DD. In alternative embodiments, the reset transistor **21** may pull the FDN to some other suitable voltage. When OFF, the reset transistor **21** has no influence on the potential of the FDN.

[0060] The source S.sub.21 of the reset transistor **21** is connected to the FDN and the gate G.sub.22 of the source-follower transistor **22**. The body B.sub.21 of the reset transistor **21** is configured to receive the second reference voltage V**2**. For example, the second reference voltage V**2** may be applied to the body B.sub.21. As above, the second reference voltage V**2** is different than and independent of the first reference voltage V**1**. Further, the second reference voltage V**2** is not tracked by the first reference voltage V**1**. The second reference voltage V**2** may, for example, be higher than the first reference voltage V**1** and/or may, for example, be 0V or ground.

[0061] The gate G.sub.22 of the source-follower transistor **22** is connected to the drain D.sub.11 of the transfer transistor **11** via the first interconnect structure **41**. As a result, the source-follower transistor **22** may monitor potential at the FDN. The drain D.sub.22 of the source-follower transistor **22** is applied with the supply voltage V.sub.DD. The source S.sub.22 of the source-follower transistor **22** is connected to the drain D.sub.23 of the row-select transistor **23**. The body B.sub.22 of the source-follower transistor **22** is configured to receive the second reference voltage V**2**. For example, the second reference voltage V**2** may be applied to the body B.sub.22.

[0062] The gate G.sub.23 of the row-select transistor **23** is configured to receive a control signal (e.g., from the ASIC **30**) to turn the row-select transistor **23** ON and OFF. When ON, the row-select transistor **23** may output an electrical signal to the ASIC **30** that represents the potential of the FDN. Otherwise, OFF, the row-select transistor **23** may isolate the pixel circuit **20** from the ASIC **30**. The drain D.sub.23 of the row-select transistor **23** is connected to the source S.sub.22 of the source-follower transistor **22**, and the source S.sub.23 of the row-select transistor **23** is connected to the ASIC **30**. In some embodiment, this connection is via is a column bus line (not shown). The body B.sub.23 of the row-select transistor **23** is configured to receive the second reference voltage V**2**. For example, the second reference voltage V**2** may be applied to the body B.sub.23.

[0063] The reset transistor **21**, the source-follower transistor **22**, and the row-select transistor **23** form the second pixel-circuit portion **20***s* that controls and processes analog signals of the of the pixel PX. The second pixel-circuit portion **20***s* is electrically connected to the pinned photodiode **10** (via the transfer transistor **11**/the first pixel-circuit portion **20***f*) and to the ASIC **30**. The second pixel-circuit portion **20***s* has a common transistor body (B.sub.21, B.sub.22, and B.sub.23) that is configured to receive the second reference voltage V**2**

[0064] Further, the reset transistor **21**, the source-follower transistor **22**, and the row-select transistor **23** are included on a substrate of the second IC die **2**. Hence, the second pixel-circuit portion **20***s* is included on the substrate. The substrate is configured to receive the second reference voltage V**2**. For example, the second reference voltage V**2** may be applied to the substrate to bias the substrate.

[0065] As noted above, the ASIC **30** includes the ADC **31** and the current source **32**. Other circuit components not shown may also be part of the ASIC **30**. In alternative embodiments, the ADC **31** may be some other suitable circuit component. Hence, the ADC **31** may more generally be referred to as an electronic device, electronic component, or the like.

[0066] The ADC **31** and the current source **32** have individual first terminals connected to the source S.sub.23 of the row-select transistor **23**. Further, the ADC **31** and the current source **32** are configured to receive the second reference voltage V**2** at individual second terminals. For example, the second reference voltage V**2** may be applied to the second terminals. The ADC **31** may be configured to convert analog signals from the pixel circuit **20** to digital signals. In some embodiments, the ADC **31** is further configured to perform pre-conversion processing on the analog signals and/or post-conversion processing on the digital signals.

[0067] The ADC **31** and the current source **32** (and more generally the ASIC **30**) are included on a substrate of the third IC die **3**. The substrate is configured to receive the second reference voltage V**2**. For example, the second reference voltage V**2** may be applied to the substrate to bias the substrate.

[0068] As noted above, the pinned photodiode **10** and the first pixel-circuit portion **20***f* are configured to receive the first reference voltage V**1**, whereas the second pixel-circuit portion **20***s* is configured to receive the second reference voltage V**2** that is independent of the first reference voltage V**1**. Further, the first reference voltage V**1** is more negative than the second reference voltage V**2**. This arrangement can achieve complete transfer of charge from the pinned photodiode **10** to the FDN via the transfer transistor **11**.

[0069] When the pinned photodiode **10** is fully depleted by a prior charge transfer, the pinned photodiode **10** has a pinning voltage V.sub.pin. In a reset mode of the pixel PX, when the transfer transistor **11** is OFF, the FDN is reset by the reset transistor **21** and the potential of the FDN is pulled up to be relatively high (e.g., the supply voltage V.sub.DD). In an integration mode of the pixel PX, when the pinned photodiode **10** is isolated from the FDN by the transfer transistor **11**, photon-induced carriers (e.g., electrons) are collected and integrated in the pinned photodiode **10**. In a charge transfer mode of the pixel PX, when the transfer transistor **11** is applied with a relatively high voltage to turn ON the transfer transistor **11**, carriers are transferred from the pinned photodiode **10** to the FDN. In a readout mode of the pixel PX, the charges stored in the FDN are converted to a voltage signal (in a relationship called “conversion gain”) by the source-follower transistor **22** and are subsequently read or processed.

[0070] Because the supply voltage V.sub.DD used to pull up the potential of the FDN is relative to the second reference voltage V**2** (e.g., 0V or ground), whereas potentials at the first IC die **1** are relative to the first reference voltage V**1** (e.g., a negative voltage), the potential to which the FDN is pulled up is effectively the supply voltage V.sub.DD increased by a magnitude of the potential difference between the first and second reference voltages V**1**, V**2**. As a result, an appropriate voltage applied to the gate G.sub.11 of the transfer transistor **11** to turn the transfer transistor **11** to a ON can cause a monotonic increase in potential from the pinned photodiode **10** to the floating diffusion node FD. This allows charges from the pinned photodiode **10** to be fully transferred to the

FDN and prevents carriers from returning to the pinned photodiode **10**.
[0071] For example, the potential of the FDN may be lower than the potential of the pinned photodiode **10** when the charges are transferred from the pinned photodiode **10** to the FDN. After the charges are fully transferred, the reverse junction voltage of the pinned photodiode **10**, which is fully depleted, may still be larger than the pinning voltage V.sub.pin. Further, the difference between the potential of the FDN and the first reference voltage V**1** is larger than the pinning voltage V.sub.pin. This guarantees charge is fully transferred.
[0072] In alternatives embodiments, the anode A.sub.10 of the pinned photodiode **10**, the body B.sub.22 of the source-follower transistor **22**, and the second terminal of the ADC **31** may be configured to respectively receive the first reference voltage V**1**, the second reference voltage V**2**, and a third reference voltage V**3** (not shown in FIG. **1**). The second reference voltage V**2** is tracked by the third reference voltage V**3** and independent of the first reference voltage V**1**. Variation of the first reference voltage V**1** is different in volume and/or in time than variation of the second reference voltage V**2**. However, variation of the second reference voltage V**2** is identical in volume and/or in time to variation of the third reference voltage V**3** because the third reference voltage V**3** tracks the second reference voltage V**2**.
[0073] In some embodiments, all terminals configured to receive the first reference voltage V**1** are electrically shorted together and/or electrically shorted to a first conductive pad, whereas all terminals configured to receive the second reference voltage V**2** are electrically shorted together and/or electrically shorted to a second conductive pad.
[0074] With reference to FIG. **2**A, a circuit diagram **200**A of some alternative embodiments of the image sensor of FIG. **1** is provided in which the pixel circuit **20** is supplied with a supply voltage V.sub.DD2 rather than supply voltage V.sub.DD. Supply voltage V.sub.DD2 is less than supply voltage V.sub.DD. For example, supply voltage V.sub.DD2 may be 1.8V, 1.5V, or some other suitable voltage, whereas supply voltage V.sub.DD may be 2.8V or some other suitable voltage. Further, the voltage difference between supply voltage V.sub.DD and supply voltage V.sub.DD2 may be equal to or substantially equal to a difference between the first reference voltage V**1** (e.g., −1.3V or some other suitable value) and the second reference voltage V**2** (e.g., 0V or ground).
[0075] As noted above, the difference between the first and second reference voltages V**1**, V**2** affectively increases the voltages at the pinned photodiode **10** and the transfer transistor **11**. Because the decrease in supply voltage from supply voltage V.sub.DD to supply voltage V.sub.DD2 is equal to or substantially equal to the difference between the first and second reference voltages V**1**, V**2**, the affective increase in voltages at the pinned photodiode **10** and the transfer transistor **11** may be sufficient to allow complete charge transfer from the pinned photodiode **10** to the FDN. For example, the potential difference between the FDN and the pinned photodiode **10** may still have a monotonic increase in potential from the pinned photodiode **10** to the FDN during charge transfer. As such, the pixel PX can fully transfer the charges (carriers) from the pinned photodiode **10** to the FDN. Further, because supply voltage V.sub.DD2 is lower than supply voltage V.sub.DD, the power consumption of the pixel circuit **20** may be reduced.
[0076] In some cases, supply voltage V.sub.DD2 means that the output swing range of the source-follower transistor **22** (or the input swing range of the ADC **31**) may be narrower than it would otherwise be with supply voltage V.sub.DD. Hence, dynamic range of the source-follower transistor **22** and the ADC **31** may be reduced. However, characteristics of the source-follower transistor **22** may be modified to at least partially regain the dynamic range. For example, the transconductance g.sub.m of the source-follower transistor **22** may be increased.
[0077] Further, the current source **32** may be configured to receive the first reference voltage V**1**, rather than the second reference voltage V**2**, to at least partially regain dynamic range. By applying the first reference voltage V**1** to the current source **32**, the voltage from the drain D.sub.22 of the source-follower transistor **22** to the second terminal of the current source **32** is effectively increased by a magnitude of the potential difference between the first and second reference voltages V**1**, V**2**.

This, in turn, increases the output swing range of the source-follower transistor **22** and the input swing range of the ADC **31**.

[0078] The current source **32** may have a relatively high effective output resistance. Further, the current source **32** may include a transistor and a decoupling capacitor. The decoupling capacitor may be connected from the gate of the transistor to the source of the transistor to stabilize the gate-to-source voltage (V.sub.GS). The relatively high effective output resistance and the stabilization of V.sub.GS may provide a constant biasing current and may minimize the noise induced by the variations of the reference voltages.

[0079] With reference to FIG. **2**B, a circuit diagram **200**B of some alternative embodiments of the image sensor of FIG. **2**A is provided in which the current source **32** has been moved to the first IC die **1**. The current source **32** may be connected to the source-follower transistor **1** through the first interconnect structure **41**. The current source **32** may provide a constant biasing current for the source-follower transistor **22**. The current source **32** is biased with the first reference voltage V**1** (same as the pinned photodiode **10**).

[0080] The current source **32** may be modified to improve the noise thereof. In such a case, special processes for manufacturing the current source **32** may be applied. That is, special processes may be applied to the manufacture of the current source **32**. The current source **32** is disposed on a substrate of the first IC die **1** independent of the substrate of the third IC die **3**. The third IC die **3**, including the ADC **31**, may be manufactured with standard processes, which are mostly fixed in a semiconductor manufacturing fab. The processes for manufacturing the first IC die **1** are relatively flexible as this IC die is for photodiodes. Thus, the current source **32** in the first IC die **1** can be optimized through the adjustment of the manufacturing processes without the constraints from the standard processes of the third IC die **3**.

[0081] With reference to FIG. **2**C, a circuit diagram **200**C of some alternative embodiments of the image sensor of FIG. **2**A is provided in which the current source **32** is configured to receive a third reference voltage V**3** rather than the first reference voltage V**1**. As noted above, supply voltage V.sub.DD2 reduces power consumption but also reduces the output swing range of the source-follower transistor **22** (or the input swing range of the ADC **31**).

[0082] In order to at least partially regain the output swing range of the source-follower transistor **22** (or the input swing range of the ADC **31**), the current source **32** may be configured to receive a third reference voltage V**3** that is less than the second reference voltage V**2**. Further, in some embodiments, the third reference voltage V**3** is higher than the first reference voltage V**1**. Because the third reference voltage V**3** is higher, the direct-current (DC) biasing for the inputs of the ADC **31** can be higher than it would otherwise be if the current source **32** was biased with the first reference voltage V**1**. This allows the DC biasing for the input of the ADC **31** to be more compatible with the designed input range of the ADC **31**.

[0083] In some embodiments, the third reference voltage V**3** and the second reference voltage V**2** are derived from the same voltage source. Although these reference voltages may be derived from the same voltage source, a voltage divider may introduce variations (or noises), such that the third reference voltage V**3** may not track the second reference voltage V**2**. In order to prevent the variations from affecting the biasing current, the current source **32** may have a relatively high effective output resistance. Further, the current source **32** may include a transistor and a decoupling capacitor connected from the gate of the transistor to the source of the transistor to stabilize the V.sub.GS.

[0084] With reference to FIG. **3**A, a circuit diagram **300**A of some embodiments of the image sensors of FIGS. **1**, **2**A, or **2**C is provided in which the ADC **31** and the current source **32** comprise individual transistors. Further, depending on embodiment of the image sensor, the pixel circuit **20** receives supply voltage V.sub.DD or supply voltage V.sub.DD2.

[0085] The ADC **31** comprises a plurality of transistors **311** (only one of which is shown) that are interconnected to implement functions of the ADC **31**. The plurality of transistors **311** comprise individual drains D.sub.311, individual sources S.sub.311, individual bodies B.sub.311, and individual gates G.sub.311. The individual bodies B.sub.311 are configured to receive the second reference voltage V**2** and are on the substrate of the third IC die **3**. Hence, the individual bodies are biased by the second reference voltage V**2** during use of the image sensor.

[0086] The current source **32** has a relatively high effective output resistance. Further, the current source **32** may include a transistor **321** and a decoupling capacitor **322** on the substrate of the third IC die **3**. The transistor **321** comprises a drain D.sub.321, a source S.sub.321, a body B.sub.321, and a gate G.sub.321. The decoupling capacitor **322** is connected from the gate G.sub.321 to the source S.sub.321 and the body B.sub.321. Further, the body B.sub.321, the source S.sub.321, and a terminal of the decoupling capacitor **322** are configured to receive the first reference voltage V**1**, the second reference voltage V**2**, or the third reference voltage V**3** depending on embodiment of the image sensor. As above, the decoupling capacitor **322** stabilizes the V.sub.GS.

[0087] With reference to FIG. **3**B, a circuit diagram **300**B of some embodiments of the image sensors of FIG. **2**B is provided in which the ADC **31** and the current source **32** comprise individual transistors. As in FIG. **3**A, the ADC **31** comprises the plurality of transistors **311** and the current source **32** comprises the transistor **321**, as well as the decoupling capacitor **322**.

[0088] With reference to FIG. **4**, a cross-sectional view **400** of some embodiments of the image sensor is provided in which the image sensor comprises a pixel array **402**. The pixel array **402** comprises a plurality of pixels PX, each being as in any one or combination of FIGS. **1**, **2**A-**2**C, **3**A, and **3**B. The first, second, and third IC dies **1**-**3** are vertically stacked with the second IC die **2** being between the first and third IC dies **1**, **3**. Further, the first IC die **1** comprises a semiconductor substrate **1***s* that is sensitive to light L**1**, whereas the second IC die **2** comprises a semiconductor substrate **2***s* that is insensitive to the light L**1** and the third IC die **3** comprises a semiconductor substate **3***s* that is insensitive to the light L**1**.

[0089] The first interconnect structure **41** is between the first and second IC dies **1**, **2** and corresponds to a bond structure. The bond structure comprises both a dielectric-to-dielectric bond and a metal-to-metal bond, whereby the bond structure may also be referred to as a hybrid bond structure. The bond structure includes a plurality of first conductive bond layers **411** and a plurality of first conductive bond vias **412** in a dielectric structure **1***d* of the first IC die **1** and further includes a plurality of second conductive bond layers **413** and a plurality of second conductive bond vias **414** in a dielectric structure **2***d* of the second IC die **2**.

[0090] The plurality of first conductive bond layers **411** are connected to the plurality of second conductive bond layers **413**. The plurality of first conductive bond layers **411** and the plurality of second conductive bond layers **413** may be bonded by metal-to-metal bonding, whereas surrounding dielectric layers **1***d* and **2***d* may be bonded by dielectric-to-dielectric bonding. Some of the plurality of first conductive bond layers **411** may be connected to the plurality of first conductive bond vias **412** while a remainder of the plurality of first conductive bond layers **411** (e.g., dummy conductive bond layers) may not be. Further, some of the plurality of second conductive bond layers **413** may be connected to the plurality of second conductive bond vias **414** while a remainder of the plurality of second conductive bond layers **413** (e.g., dummy conductive bond layer) may not be.

[0091] The first IC die **1** includes a plurality of conductive wire levels M**11**, M**12** and one or more conductive via levels V**11** in the dielectric structure **1***d* of the first IC die **1**. Although FIG. **4** shows two conductive wire levels and one conductive via level, the first IC die **1** may include more conductive wire levels and/or more conductive via levels. Conductive wire level M**11** is connected to conductive wire level M**12** by via level V**11**, and conductive wire level M**12** is connected to the first conductive bond vias **412**. The first IC die **1** further includes a plurality of conductive contacts C**11** in the dielectric structure **1***d*.

[0092] The plurality of conductive contacts C**11** connect conductive wire level M**11** to a bulk of the semiconductor substrate **1**s of the first IC die **1**. Further, the plurality of conductive contacts C**11** connect conductive wire level M**11** to a plurality of transfer transistors **11** (more generally, first pixel-circuit portions). The plurality of transfer transistors **11** are partially shown and are associated with a plurality of pinned photodiodes **10** in the semiconductor substrate **1***s* of the first IC die **1**. In some embodiments, there is a one-to-one correspondence between the plurality of transfer transistors **11** and the plurality of pinned photodiodes **10**.

[0093] The second IC die **2** includes a plurality of conductive wire levels M**21**, M**22** and one or more conductive via levels V**21** in the dielectric structure **2***d* of the second IC die **2**. Although FIG. **4** shows two conductive wire levels and one conductive via level, the second IC die **2** may include more conductive wire levels and/or more conductive via levels. Conductive wire level M**21** is connected to conductive wire level M**22** by via level V**21**, and conductive wire level M**22** is connected to the second conductive bond vias **414**. The second IC die **2** further includes a plurality of conductive contacts C**21** in the dielectric structure **2***d*.

[0094] The plurality of conductive contacts C**21** connect conductive wire level M**21** to a bulk of the semiconductor substrate **2**s of the second IC die **2**. Further, the plurality of conductive contacts C**21** connect conductive wire level M**21** to a plurality of second pixel-circuit portions **20***s*. In some embodiments, there is a one-to-one or one-to-many correspondence between the second pixel-circuit portions **20***s* and the plurality of pinned photodiodes **10**. The second pixel-circuit portions **20***s* comprise a plurality of transistors **20***t*, which correspond to the reset transistors **21**, the source-follower transistors **22**, and the row-select transistors **23** previously discussed. The plurality of transistors **20***t* may, for example, each be connected to conductive wire level M**21** by the plurality of conductive contacts C**21**.

[0095] The plurality of pinned photodiodes **10**, the plurality of first pixel-circuit portions **20***f* (e.g., the plurality of transfer transistors **11**), and the plurality of second pixel-circuit portions **20***s* are interconnected by the wire and via levels (e.g., conductive wire level M**21**, via level V**11**, etc.) and the first interconnect structure **41** to form the pixel array **402**. Each pixel PX includes a pinned photodiode **10**, a first pixel-circuit portion **20***f*, and a second pixel-circuit portion **20***s* and may, for example, be as in any one or combination of FIGS. **1**, **2**A-**2**C, **3**A, and **3**B.

[0096] The second interconnect structure **42** corresponds to a plurality of conductive through vias. The plurality of conductive through vias extend through the semiconductor substrate **2***s* of the second IC die **2** and are separated from the semiconductor substrate **2***s* by dielectric liners **421**. The conductive through vias may be copper, gold, aluminum, or the like.

[0097] The third IC die **3** includes a plurality of conductive wire levels M**31**, M**32**, M**33** and a plurality of conductive via levels V**31**, V**32** in the dielectric structure **3***d* of the third IC die **3**. Although FIG. **4** shows three conductive wire levels and two conductive via levels, the third IC die **3** may include more conductive wire levels and/or more conductive via levels. Conductive wire level M**31** is connected to conductive wire level M**32** by via level V**31**, conductive wire level M**32** is connected to conductive wire level M**33** by via level V**32**, and conductive wire level M**33** is connected to the plurality of pixel circuits **20** by the second interconnect structure **42**. The third IC die **3** further includes a plurality of conductive contacts C**31**.

[0098] The plurality of conductive contacts C**31** are in the dielectric structure **3***d* of the third IC die **3** and connect conductive wire level M**31** to a bulk of the semiconductor substrate **3***s* of the third IC die **3**. Further, the plurality of conductive contacts C**31** connect conductive wire level M**31** to an ASIC **30**. The ASIC **30** comprises individual transistors **30***t*. The individual transistors **30***t* form the ADC **31** and/or the current source **32** previously discussed and may each be connected to conductive wire level M**31** by the plurality of conductive contacts C**31**.

[0099] The first IC die **1** includes a plurality of conductive pads **51**, including a first conductive pad **51***a* and a second conductive pad **51***b*, at a top of the first IC die **1**, laterally adjacent to the semiconductor substrate **1**s of the first IC die **1**. The first IC die **1** includes one or more conductive pad vias **52***a* connecting the first conductive pad **51***a* with conductive wire level M**11** and further includes one or more conductive pad vias **52***b* connecting the second conductive pad **51***b* with conductive wire level M**11**. The plurality of conductive pads **51** may, for example, be or comprise aluminum copper, copper, some other suitable conductive material, or a combination of the foregoing. Further, the plurality of conductive pads **51** may be connected to wire bonds (not shown) which may electrically connect a power source.

[0100] As previously discussed, the pinned photodiodes **10** and the transfer transistors **11** are configured to receive the first reference voltage V**1**. In FIG. **4**, the image sensor is configured to receive the first reference voltage V**1** at the first conductive pad **51***a*. Conductive features (e.g., wires, vias, contacts, etc.) of the first IC die **1** pass the first reference voltage V**1** from the first conductive pad **51***a* to the semiconductor substrate **1**s of the first IC die **1**. Further, the semiconductor substrate **1**s forms one or more conductive paths to pass the first reference voltage V**1** to the pinned photodiodes **10** (e.g., anodes A.sub.10 of the pinned photodiodes **10**) and the transfer transistors **11** (e.g., bodies B.sub.11 of transfer transistors **11**).

[0101] Also, as previously discussed, the second pixel-circuit portions **20***s* and the ASIC **30** are configured to receive the second reference voltage V**2**. In FIG. **4**, the image sensor is configured to receive the second reference voltage V**2** at the second conductive pad **51***b*. Conductive features (e.g., wires, vias, contacts, interconnect structures, etc.) in the first, second, and third IC dies **1-3** pass the second reference voltage V**2** from the second conductive pad **51***b*to the semiconductor substrate **2***s* of the second IC die **2** and to the semiconductor substrate **3***s* of the third IC die **2**. The semiconductor substrate **2***s* of the second IC die **2** forms one or more conductive paths to pass the second reference voltage V**2** to the transistors **20***t* of the second pixel-circuit portions **20***s* (e.g., bodies of the transistors **20***t*). Similarly, the semiconductor substrate **3***s* of the third IC die **2** forms one or more conductive paths to pass the second reference voltage V**2** to the transistors **30***t* of the ASIC **30** (e.g., bodies of the transistors **30***t*).

[0102] Because the second and third IC dies **2**, **3** have a common reference or ground level, the noise induced by variations of the ground level can be reduced and thus the noise at the ASIC **30** can be improved. For example, noise at an ADC **31** of the ASIC **30** (see, e.g., FIG. **1**) may be reduced and SNR of the ADC **31** may be improved.

[0103] With continued reference to FIG. **4**, the first IC die **1** may include a plurality of color filters **53**, a plurality of microlenses **54**, and an anti-reflection layer **55**. The microlenses **54** are respectively on the color filters **53** and are aligned with the pinned photodiodes **10** and the color filters **53**. Further, the microlenses **54** have a curved (e.g., convex) surface that focuses the light L**1** respectively on the pinned photodiodes **10**.

[0104] The color filters **53** are on the anti-reflection layer **55**, which is between the color filters **53** and the semiconductor substrate **1***s* of the first IC die **1**. The anti-reflection layer **55** is configured to minimize reflection of incident light and allows more light to reach the pinned photodiodes **10**. In some embodiments, the anti-reflection layer **55** may be or comprise, for example, oxide, nitride, high-k dielectric material, or the like. The high-k dielectric material may, for example, be or comprise aluminum oxide (AlO), tantalum oxide (TaO), hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium aluminum oxide (HfAlO), or hafnium tantalum oxide (HMO), or a combination thereof.

[0105] During back-side illumination (BSI) of the image sensor, the light L**1** is incident on the back of the first IC die **1** and passes, in sequence, through the microlenses **54**, the color filters **53**, and the anti-reflection layer **55**. The light L**1** then passes into the semiconductor substrate **1**s of the first IC die **1** to reach the pinned photodiodes **10**. In alternative embodiments, the first IC die **1** is vertically flipped while the second and third IC dies **2**, **3**, remain as is. In such embodiments, the back of the first IC die **1** is bonded to the second IC die **2** and the first IC die **1** receives the light L**1** at a front of the first IC die **1** for front-side illumination (FSI).

[0106] With reference to FIG. **5**, a plan view **500** of some embodiments of the image sensor of FIG. **4** is provided. The cross-sectional view **400** of FIG. **4** may, for example, be taken along line A-A′ in FIG. **4**. The image sensor comprises the pixel array **402**, which comprises a plurality of pixels PX

in a plurality of rows and a plurality of columns. Further, each pixel PX is configured as in any one or combination of FIGS. **1**, **2**A-**2**C, **3**A, and **3**B.
[0107] A plurality of conductive pads **51** are spaced in a ring-shaped path around the pixel array **402**. The plurality of conductive pads **51** include the first conductive pad **51***a*, the second conductive pad **51***b*, and a third conductive pad **51***c*. The first and second conductive pads **51***a*, **51***b* are configured to respectively receive the first reference voltage V**1** and the second reference voltage V**2**. As above, pinned photodiodes **10** of the pixels PX and transfer transistors **11** of the pixels PX are configured to receive the first reference voltage V**1** from the first conductive pad **51***a*, whereas the pixel circuits **20** of the pixels PX are configured to receive the second reference voltage V**2** from the second conductive pad **51***b*.
[0108] In some embodiments, anodes of the pinned photodiodes **10** and bodies of the transfer transistors **11** are electrically shorted to the first conductive pad **51***a* by conductive features in the first IC die **1**. In some embodiments, bodies of the transistors **20***t* of the second pixel-circuit portions **20***s* and bodies of the transistors **30***t* forming the ASIC **30** are electrically shorted to the second conductive pad **51***b* by conductive features in the first, second, and third IC dies **1-3**.
[0109] In some embodiments, depending upon the configuration of the pixels PX in FIGS. **4** and **5**, the third conductive pad **51***c* is configured to receive the third reference voltage V**3**. Further, the current source **32** of the ASIC **30** (see, e.g., FIG. **2**C) is configured to receive the third reference voltage V**3** from the third conductive pad **51***c*. In some embodiments, a transistor body of the current source **32** is electrically shorted to the third conductive pad **51***c* by conductive features in the first, second, and third IC dies **1-3**.
[0110] With reference to FIG. **6**, a circuit diagram **600** of some embodiments of an image sensor that spans two IC dies, according to aspects of the present disclosure, is provided. The image sensor is similar to the image sensor of FIG. **1** and includes the first IC die **1** and the second IC die **2** as in FIG. **1**. However, in contrast with the image sensor of FIG. **1**, the third IC die **3** is omitted and the ASIC **30** is in the second IC die **2**.
[0111] The pixel circuit **20** is in the second IC die **2** and is configured as described with regard to FIG. **1**. Individual transistors of the second pixel-circuit portion **20***s* have bodies configured to receive the second reference voltage V**2**. Further, the ASIC **30** is in the second IC die **2** and is configured as described with regard to FIG. **1**. The ADC **31** of the ASIC **30** and the current source **32** of the ASIC **30** are configured to receive the second reference voltage V**2**. Because the second pixel-circuit portion **20***s* and the ASIC **30** have the same reference voltage, variations in reference voltage are the same in the second pixel-circuit portion **20***s* and the ASIC **30**. As a result, noise is reduced and the SNR of the ADC **31** is not negatively impacted.
[0112] The pinned photodiode **10** and the transfer transistor **11** are in the first IC die **1**, and the transfer transistor **11** is electrically coupled to the pixel circuit **20** through the first interconnect structure **41**. Further, the pinned photodiode **10** and the transfer transistor **11** are configured to receive the first reference voltage V**1**, which is independent of the second reference voltage V**2**. As above, the first reference voltage V**1** may be more negative than the second reference voltage V**2** to effectively increase supply voltage at the first IC die **1** by a magnitude of the potential difference between the first and second reference voltages V**1**, V**2**. This increases the FWC of the pinned photodiode **10** and allows complete charge transfer of photon-induced carriers out of the pinned photodiode **10**.
[0113] With reference to FIG. **7**A, a circuit diagram **700**A of some alternative embodiments of the image sensor of FIG. **6** is provided in which the pixel circuit **20** is supplied with a supply voltage V.sub.DD2 rather than supply voltage V.sub.DD. Further, supply voltage V.sub.DD2 is less than supply voltage V.sub.DD. As a result, the output swing range of the source-follower transistor **22** (or the input swing range of the ADC **31**) may be narrower. In order to regain this loss in range, the current source **32** may be configured to receive the first reference voltage V**1**.
[0114] By applying the first reference voltage V**1** to the current source **32**, the voltage from the

drain D.sub.22 of the source-follower transistor **22** to the second terminal of the current source **32** is effectively increased by a magnitude of the potential difference between the first and second reference voltages V**1**, V**2**. This, in turn, increases the output swing range of the source-follower transistor **22** and the input swing range of the ADC **31**.
[0115] The current source **32** may have a relatively high effective output resistance. Further, the current source **32** may include a transistor and a decoupling capacitor connected from the gate of the transistor to the source of the transistor to stabilize the V.sub.GS. The relatively high effective output resistance and the stabilization of V.sub.GS may provide a constant biasing current and may minimize the noise induced by the variations of the reference voltages.
[0116] With reference to FIG. **7**B, a circuit diagram **700**B of some alternative embodiments of the image sensor of FIG. **7**A is provided in which the current source **32** has been moved to the first IC die **1**. The current source **32** may be connected to the source-follower transistor **1** through the first interconnect structure **41**. The current source **32** may provide a constant biasing current for the source-follower transistor **22** and is biased with the first reference voltage V**1** (same as the pinned photodiode **10**).
[0117] The current source **32** may be modified to improve the noise thereof. For example, special processes for manufacturing the current source **32** may be applied. The current source **32** is in a substrate of the first IC die **1** independent of the substrate of the third IC die **3**. The third IC die **3**, including the ADC **31**, may be manufactured with standard processes, which are mostly fixed in a semiconductor manufacturing fab. The processes for manufacturing the first IC die **1** are relatively flexible as this IC die is for photodiodes. Thus, the current source **32** in the first IC die **1** can be optimized through the adjustment of the manufacturing processes without the constraints from the standard processes of the third IC die **3**.
[0118] With reference to FIG. **7**C, a circuit diagram **700**C of some alternative embodiments of the image sensor of FIG. **7**A is provided in which the current source **32** is configured to receive a third reference voltage V**3** rather than the first reference voltage V**1**. As noted above, supply voltage V.sub.DD2 reduces power consumption but also reduces the output swing range of the source-follower transistor **22** (or the input swing range of the ADC **31**).
[0119] In order to at least partially regain the output swing range of the source-follower transistor **22** (or the input swing range of the ADC **31**), the current source **32** can be configured to receive a third reference voltage V**3** that is less than the second reference voltage V**2**. Further, in some embodiments, the third reference voltage V**3** is higher than the first reference voltage V**1**. Because the third reference voltage V**3** is higher, the DC biasing for the inputs of the ADC **31** can be higher than they would otherwise be if the current source **32** was biased with the first reference voltage V**1**. This allows the DC biasing for the input of the ADC **31** to be more compatible with the designed input range of the ADC **31**.
[0120] In some embodiments, the third reference voltage V**3** and the second reference voltage V**2** are derived from the same voltage source. Although these reference voltages may be derived from the same voltage source, a voltage divider may introduce variations (or noises), such that the third reference voltage V**3** may not track the second reference voltage V**2**. In order to prevent the variations from affecting the biasing current, the current source **32** may have a relatively high effective output resistance. Further, the current source **32** may include a transistor and a decoupling capacitor connected from the gate of the transistor to the source of the transistor to stabilize the V.sub.GS.
[0121] With reference to FIG. **8**A, a circuit diagram **800**A of some embodiments of the image sensors of FIGS. **6**, **7**A, or **7**C is provided in which the ADC **31** and the current source **32** comprise individual transistors. Further, depending on embodiment of the image sensor, the pixel circuit **20** receives supply voltage V.sub.DD or supply voltage V.sub.DD2.
[0122] The ADC **31** comprises a plurality of transistors **311** that are interconnected to implement functions of the ADC **31** and that have individual bodies B.sub.311 configured to receive the

second reference voltage V**2**. The current source **32** comprises a transistor **321** and a decoupling capacitor **322**. The decoupling capacitor **322** is connected from the gate G.sub.321 of the transistor **321** to the source S.sub.321 of the transistor **321** and the body B.sub.321 of the transistor **321**. Further, the body B.sub.321, the source S.sub.321, and a terminal of the decoupling capacitor **322** are configured to receive the first reference voltage V**1**, the second reference voltage V**2**, or the third reference voltage V**3** depending on embodiment of the image sensor.

[0123] With reference to FIG. **8**B, a circuit diagram **800**B of some embodiments of the image sensors of FIG. **7**B is provided in which the ADC **31** and the current source **32** comprise individual transistors. As in FIG. **8**A, the ADC **31** comprises the plurality of transistors **311** and the current source **32** comprises the transistor **321**, as well as the decoupling capacitor **322**.

[0124] With reference to FIG. **9**, a cross-sectional view **900** of some embodiments of the image sensor is provided in which the image sensor comprises a pixel array **402**. The pixel array **402** comprises a plurality of pixels PX, each being as in any one of combination of FIGS. **6**, **7**A-**7**C, **8**A, and **8**B. The image sensor is as in FIG. **4**, except that the third IC die **3** has been omitted. Further, the ASIC **30** has been moved to the second IC die **2**, laterally offset from the plurality of pixel circuits **20**. As such, the ASIC **30** and the plurality of pixel circuits **20** share the semiconductor substrate **2***s* of the second IC die **2**. This may reduce the cost to manufacture the image sensor and may improve yields, as compared to the image sensor of FIG. **4**.

[0125] As previously discussed, the pinned photodiodes **10** and the transfer transistors **11** (more generally, the first pixel-circuit portions **20***f*) are configured to receive the first reference voltage V**1**. For example, as illustrated in the cross-sectional view **900**, the image sensor is configured to receive the first reference voltage V**1** at the first conductive pad **51***a*. Conductive features (e.g., wires, vias, contacts, etc.) of the first IC die **1** pass the first reference voltage V**1** from the first conductive pad **51***a* to the semiconductor substrate **1***s* of the first IC die **1**. Further, the semiconductor substrate **1***s* forms one or more conductive paths to pass the first reference voltage V**1** to the pinned photodiodes **10** (e.g., anodes A.sub.10 of the pinned photodiodes **10**) and the transfer transistors **11** (e.g., bodies B.sub.11 of transfer transistors **11**).

[0126] Also, as previously discussed, the second pixel-circuit portions **20***f* and the ASIC **30** are configured to receive the second reference voltage V**2**. For example, as illustrated in the cross-sectional view **900**, the image sensor is configured to receive the second reference voltage V**2** at the second conductive pad **51***b*. Conductive features (e.g., wires, vias, contacts, interconnect structures, etc.) in the first and second IC dies **1**, **2** pass the second reference voltage V**2** from the second conductive pad **51***b* to the semiconductor substrate **2***s* of the second IC die **2**. The semiconductor substrate **2***s* of the second IC die **2** forms one or more conductive paths to pass the second reference voltage V**2** to bodies of the transistors **20***t* of the second pixel-circuit portions **20***f* and bodies of the transistors **30***t* of the ASIC **30**.

[0127] Because the second pixel-circuit portions **20***f* and the ASIC **30** have a common reference or ground level, the noise induced by variations of the ground level can be reduced and thus the noise at the ASIC **30** can be improved. For example, noise at an ADC **31** of the ASIC **30** (see, e.g., FIG. **6**) may be reduced and SNR of the ADC **31** may be improved.

[0128] With continued reference to FIG. **9**, the pixel circuits **20** and the pinned photodiodes **10** may overlap vertically. Further, the ASIC **30** may be free from vertical overlap with the pinned photodiodes **10**. Instead, the ASIC **30** may vertically overlap with a dummy region DM of the first IC die **1** that is not used for the conversion of light (e.g., photon) to carriers (e.g. electron). The dummy region DM may include dummy conductive patterns to avoid a large difference in pattern density between the pinned photodiodes **10** and the dummy region DM. Further, in some embodiments, the dummy region DM may include a capacitor, resistor, or transistor that electrically connected to the pinned photodiodes **10**.

[0129] With reference to FIG. **10**, a plan view **1000** of some embodiments of the image sensor of FIG. **9** is provided. The cross-sectional view **900** of FIG. **9** may, for example, be taken along line

B-B′ in FIG. **10**. The image sensor comprises a pixel array **402** and an ASIC **30** that laterally border each other. The pixel array **402** comprises a plurality of pixels PX in a plurality rows and a plurality of columns. Further, each pixel PX is configured as in any one or combination of FIGS. **6**, **7**A-**7**C, **8**A, and **8**B.

[0130] A plurality of conductive pads **51** are spaced in a ring-shaped path around the pixel array **402** and the ASIC **30** and include the first conductive pad **51***a*, the second conductive pad **51***b*, and a third conductive pad **51***c*. The first and second conductive pads **51***a*, **51***b* are configured to respectively receive the first reference voltage V**1** and the second reference voltage V**2**. As above, pinned photodiodes **10** of the pixels PX and transfer transistors **11** of the pixels PX are configured to receive the first reference voltage V**1** from the first conductive pad **51***a*, whereas the ASIC **30** and the pixel circuits **20** of the pixels PX are configured to receive the second reference voltage V**2** from the second conductive pad **51***b*.

[0131] In some embodiments, anodes of the pinned photodiodes **10** and bodies of the transfer transistors **11** are electrically shorted to the first conductive pad **51***a* by conductive features in the first IC die **1**. Further, in some embodiments, bodies of the transistors **20***t* of the second pixel-circuit portions **20***f* and bodies of the transistors **30***t* of the ASIC **30** are electrically shorted to the second conductive pad **51***b* by conductive features in the first and second IC dies **1**, **2**.

[0132] In some embodiments, depending upon the configuration of the pixels PX in FIGS. **9** and **10**, the third conductive pad **51***c* is configured to receive the third reference voltage V**3**. Further, the current source **32** of the ASIC **30** (see, e.g., FIG. **7**C) is configured to receive the third reference voltage V**3** from the third conductive pad **51***c*. In some embodiments, a transistor body of the current source **32** is electrically shorted to the third conductive pad **51***c* by conductive features in the first and second IC dies **1**, **2**.

[0133] As seen above, and similar to FIGS. **1**, **2**A-**2**C, **3**A, **3**B, **4-6**, **7**A-**7**C, **8**A, **8**B, **9** and **10** illustrate various different reference voltages. In some embodiments, all terminals configured to receive the first reference voltage V**1** are electrically shorted together and/or electrically shorted to a first conductive pad, all terminals configured to receive the second reference voltage V**2** are electrically shorted together and/or electrically shorted to a second conductive pad, and all terminals configured to receive the third reference voltage V**3** are electrically shorted together and/or electrically shorted to a third conductive pad. Such electrical shorting may, for example, be by conductive wires, vias, contacts, and the like of the image sensor.

[0134] With reference to FIG. **11**A, a block diagram **1100**A of some embodiments of an image sensor comprising a plurality of pixels PX, each as in any one or combination of FIGS. **1**, **2**A-**2**C, **3**A, **3**B, **4-6**, **7**A-**7**C, **8**A, **8**B, **9** and **10**, is provided. The plurality of pixels PX are in a plurality of rows and a plurality or columns to form a pixel array **402**. For example, at least 3 rows and at least 3 columns are illustrated. However, there may be more or less rows and/or more or less columns in alternative embodiments.

[0135] The plurality of pixels PX comprise individual pinned photodiodes **10** and individual pixel circuits **20**, which include first pixel-circuit portions **20***f* (e.g., transfer transistors **11**) and second pixel-circuit portions **20***s*. Further, the plurality of pixels PX share the ASIC **30** and are each electrically coupled to the ASIC **30**. Therefore, there may be only one ASIC **30** in at least some embodiments. As above, the pinned photodiodes **10** and the transfer transistors **11** are in the first IC die **1** with the first reference voltage V**1**, and the second pixel-circuit portions are in the second IC die **2** with the second reference voltage V**2**. Further, the ASIC **30** is at least partially in a second or third IC die **2**, **3** with the second reference voltage V**2**.

[0136] With reference to FIG. **11**B, a circuit diagram **1100**B of a column in the pixel array **402** of FIG. **11**A is provided. Pixels PX in the column are individually connected to a conductive line **1104**. Said connection may, for example, be via sources of row-select transistors **23** in the individual pixels PX. The conductive line **1104** extends to the ASIC **30** and electrically couples with an ADC **31** and a current source **32**, which are as in any one or combination of FIGS. **1**, **2**A-

2C, 3A, 3B, 4-6, 7A-7C, 9A, 9B, and 10.

[0137] In some embodiments, the column illustrated by FIG. **11**B is representative of each individual column in the pixel array **402** of FIG. **11**A. Further, in some embodiments, the conductive line **1104**, the ADC **31**, and the current source **32** repeat for each individual column of the pixel array **402**, such that the ASIC **30** may include an ADC and a current source for each individual column of the pixel array **402**.

[0138] In view of the foregoing, at least some embodiments of the present disclosure relate to an image sensor including a first group of devices with at least one pinned photodiode and at least one transfer transistor, a second group of devices with at least one pixel circuit, and a third group of devices with an electronic device. The electronic device may, for example, correspond to the ADC **31** or some other suitable electronic device.

[0139] The first group of devices is connected to the second group of devices through a first interconnect structure. The pixel circuits of the second group of devices control at least one pinned photodiode and at least one transfer transistor of the first group of devices. The electronic device processes signals from the second group of devices. The first group of devices is biased with a negative reference voltage. The second group of devices and the electronic device are biased with the same reference voltage, providing the second group of devices and the electronic device with the same ground level. Hence, variations in ground will not induce noise and, at least where the electronic device is an ADC, the SNR of the ADC is improved

[0140] Further, the negative reference voltage applied to the first group of devices is more negative than the reference voltage that is applied to the second group of devices. In the reset mode of the image sensor, the at least one pinned photodiode can be sufficiently reset by the at least one pixel circuits and, in the charge transfer mode of the image sensor, photon-induced carriers in the at least one pinned photodiode can be completely transferred to at least one FDN.

[0141] While the present disclosure has thus far focused on image sensors, it is to be appreciated that the aforementioned concepts are applicable to other device types. For example, circuit components can be shifted, and/or additional voltages can be added to other device types, to reduce noise and/or enhance performance. Such other device types include, for example, Microelectromechanical Systems (MEMS) microphones, biosensors, three-dimensional (3D) memory, 3D fabric, photonics, and so on.

[0142] With reference to FIG. **12**, a circuit diagram **1200** of some embodiments of a biosensor that spans three IC dies, according to aspects of the present disclosure, is provided. The biosensor is similar to the image sensor of FIG. **1** in that it includes the first IC die **1**, the second IC die **2**, and the third IC die **3**. However, as seen below, there are a number of differences to account for the change in sensor type.

[0143] The first and second IC dies **1**, **2** are electrically coupled and bonded together by the first interconnect structure **41**, whereas the second and third IC die **2**, **3** are electrically coupled together by an internal wire bond structure **43**. Further, the pixel PX spans the first and second IC dies **1**, **2** and comprises a bio field-effect transistor (BioFET) **12**, a reference electrode **13**, and a pixel circuit **20**. The BioFET **12** and the reference electrode **13** are in the first IC die **1**, and the pixel circuit **20** is entirely within the second IC die **2**. The BioFET **12** may, for example, be an ion-sensitive field-effect transistor (ISFET) or some other suitable type of BioFET.

[0144] The BioFET **12** is configured to sense a variety of bioentities present in a fluid proximate a channel C.sub.12 of the BioFET **12**. Such bioentities may, for example, include molecules like DNA, proteins, cells, and so on. In some embodiments, the sensing mechanism relies on detecting changes in ion concentration near the channel C.sub.12. When a bioentity binds to a specialized bio-sensing film at the channel C.sub.12, it triggers the release of ions into the fluid. For example, when a DNA segment binds, it may release hydrogen ions (H+), altering the local pH. The change in ion concentration impacts the conductivity of the channel C.sub.12, thereby providing a measurable output signal that is proportional to the amount of bound bioentity.

[0145] The reference electrode **13** is configured to receive a reference voltage V.sub.ref and create a stable electric field within the channel C.sub.12 of the BioFET **12**. As a result, the reference electrode **13** acts as a gate electrode to control channel conductivity. This enables accurate detection of subtle shifts caused by bioentities and tuning of the BioFET sensitivity.
[0146] The BioFET **12** has the channel C.sub.12, a front gate formed by the reference electrode **13**, and a back gate BG.sub.12 electrically coupled to the pixel circuit **20**. Further, the BioFET **12** has a first terminal electrically coupled to an input of the pixel circuit **20** and a second terminal configured to receive the first reference voltage V**1**. As seen hereafter, the first reference voltage V**1** may be negative to increase a voltage across the BioFET **12** during sensing, thereby increasing sensitivity and output range of the BioFET **12**. The first terminal may, for example, correspond to a drain D.sub.12 of the BioFET **12**, and the second terminal may, for example, correspond to a source S.sub.12 of the BioFET **12**, or vice versa.
[0147] The pixel circuit **20** is electrically coupled to the first terminal (e.g., output) of the BioFET **12** via the first interconnect structure **41** and is configured to facilitate readout and operation of the BioFET **12**. The pixel circuit **20** comprises a plurality of pixel transistors that have individual transistor bodies (e.g., body B.sub.24 and body B.sub.25) electrically coupled together (e.g., via a substrate of the second IC die **2**) and that are configured to receive a second reference voltage V**2** greater than the first reference voltage V**1**. The plurality of pixel transistors include a support transistor **24** and a select transistor **25**.
[0148] The support transistor **24** is configured to selectively electrically couple a back gate BG.sub.12 of the BioFET **12** to a fourth voltage V**4** and/or a fifth voltage V**5**. The fourth voltage V**4** and/or a fifth voltage V**5** may, for example, be received from the ASIC **30**.
[0149] In some embodiments, the fourth voltage V**4** is applied between sensing cycles to repel ions and/or charges that have accumulated at the BioFET **12** from previous sensing cycles. Such ions and/or charges influence the conductance of the channel C.sub.12 and hence induce skew the measurements by the BioFET **12**. The fourth voltage V**4** may, for example, be positive if the ions and/or charge are positive and may, for example, be negative if the ions and/or charge are negative. In some embodiments, the fifth voltage V**5** is applied during sensing to attract ions to the channel C.sub.12 to increase sensitivity of the BioFET **12**. For example, the fifth voltage V**5** may be positive if the ions are negative and may be negative if the ions are positive.
[0150] In some embodiments, the fourth voltage V**4** is different than the first voltage V**1** and/or the second voltage V**2**, and/or the fifth voltage V**5** is different than the first voltage V**1** and/or the second voltage V**2**. In some embodiments, the fourth voltage V**4** or the fifth voltage V**5** is equal to or about equal to the first voltage V**1**. In some embodiments, the fourth voltage V**4** is greater than the first voltage V**2**, greater than the second voltage V**2**, between the first voltage V**1** and the second voltage V**2**, or less than the first voltage V**1**. Further, in some embodiments, the fourth voltage V**5** is greater than the first voltage V**2**, greater than the second voltage V**2**, between the first voltage V**1** and the second voltage V**2**, or less than the first voltage V**1**. In some embodiments, terminals sharing voltages in any figure herein may be electrically shorted together by conductive features of the semiconductor structure.
[0151] The select transistor **25** is configured to selectively electrically couple the first terminal (e.g., an output) of the BioFET **12** to an input of the ASIC **30**. The pixel PX may, for example, be one of many pixels electrically coupled to the input of the ASIC **30**, whereby the select transistor **25** allows the pixels to be readout sequentially over time using the same signal line and circuitry (e.g., amplifier, ADC, etc.) within the ASIC **30**.
[0152] The ASIC **30** is in the third IC die **3** and electrically coupled to the pixel circuit **20** by the internal wire bond structure **43**. The ASIC **30** may, for example, provide control over operation of the pixel PX and/or may, for example, perform analog-to-digital conversion (ADC), signal processing, and the like on an output of the pixel circuit **20**. The ASIC **30** comprises an ADC **31**, a transimpedance amplifier (TIA) **33**, and a controller **34**. The ASIC **30**, the TIA **33**, and the

controller **34** (more generally the ASIC **30**) are configured to receive the second reference voltage V**2**. For example, the ADC **31**, the TIA **33**, and the controller **34** may comprise transistors with bodies configured to receive the second reference voltage V**2**.

[0153] The TIA **33** is electrically coupled from an output of the pixel circuit **20** to an input of the ADC **31**. The TIA **33** is configured to convert a variable current produced by the BioFET **12** during sensing into a voltage, and the ADC **31** is configured to convert the voltage into a digital signal that can undergo digital processing. The controller **34** is configured to provide control signals to the pixel PX to change the pixel between sensing and resetting. For example, the controller **34** may turn the support transistor **24** to an ON state, turn the select transistor **25** to an OFF state, and provide the fourth voltage V**4** to the support transistor **24** while resetting the BioFET **12** between sensing cycles. As another example, the controller **34** may turn the support transistor **24** to an ON, turn the select transistor **25** to an ON state, and provide the fifth voltage V**5** to the support transistor **24** during a sensing cycle with the BioFET **12**.

[0154] In some embodiments, the BioFET **12** may be generalized as a sensor, a functional device, or the like. In some embodiments, the pixel circuit **20** may be generalized as an interface circuit for the BioFET **12**. In some embodiments, the ASIC **30** may be generalized as a control and/or processing circuit for the pixel circuit **20** and/or the BioFET **12**.

[0155] A reference voltage (more generally any voltage) is subject to (random) variations. Such variations may, for example, correspond to stray voltages and/or may, for example, arise from the resistance of bond wires, conductive traces, etc. in the IC dies. With different reference voltages, the variations in one reference voltage are different than, and hence don't track, the variations in the one or more other reference voltages. This non-tracking may lead to noise that negatively affects the performance (e.g., the SNR) of the ADC **31**.

[0156] Because the pixel circuit **20**, the ADC **31**, and the TIA **33** are configured to receive the second reference voltage V**2** (e.g., the same reference voltage), these circuit components have the same reference voltage. Hence, the ADC **31** has the same reference-voltage variations as the pixel circuit **20** and the TIA **33**. This prevents noise induced by different reference voltages respectively at the pixel circuit **20** and the ASIC **30** and enhances the performance.

[0157] In some embodiments, the first reference voltage V**1** is negative, whereas the second reference voltage V**2** is ground or otherwise greater than the first reference voltage V**1**. As a result, the voltage across the BioFET **12** is relative to the first reference voltage V**1** rather than the second reference voltage V**2**. This effectively increases this voltages by a magnitude of the potential difference between the first and second reference voltages V**1**, V**2**.

[0158] The affective increase in voltage at the BioFET **12** allows the supply voltage to be reduced and/or performance of the biosensor to be enhanced. For example, the reduced supply voltage reduces power consumption and may, for example, coincide with scaling down of the biosensor. As another example, the increased voltage increases the sensitivity and output range of the BioFET **12** to enhance the performance of the biosensor.

[0159] With continued reference to FIG. **12**, the support transistor **24** includes a gate G.sub.24, a source S.sub.24, a drain D.sub.24, and a body B.sub.24. The body B.sub.24 is configured to receive the second voltage V**2**, and the gate G.sub.24 is configured to receive a control signal (e.g., from the ASIC **30**) to selectively change the support transistor **24** between an ON state and an OFF state. The source S.sub.24 is electrically coupled to the back gate BG.sub.12 of the BioFET **12** and the drain D.sub.24 is electrically coupled to the controller **34** of the ASIC **30**. Note that the connection is broken for ease of illustrations and both terminals labeled “A” are the same.

[0160] The select transistor **25** includes a gate G.sub.25, a source S.sub.25, a drain D.sub.25, and a body B.sub.25. The body B.sub.25 is configured to receive the second voltage V**2**, whereas the gate G.sub.25 is configured to receive a control signal (e.g., from the ASIC **30**) to selectively change the select transistor **25** between an ON state and an OFF state. The source S.sub.25 is electrically coupled to an input of the TIA **33**, and the drain D.sub.24 is electrically coupled to the second terminal (e.g., output) of the BioFET **12**. In some embodiments, the source S.sub.25 is electrically coupled to the input of the TIA **33** via is a bus line (not shown) shared with other pixels.

[0161] In some embodiments, the support transistor **24** and the select transistor **25** are N-channel field-effect transistors. In alternative embodiments, the support transistor **24** and the select transistor **25** are P-channel field-effect transistors. In at least some of such alternative embodiments, the electrical couplings of the sources S.sub.24, S.sub.25 and the electrical couplings of the drains D.sub.24, D.sub.25 are reversed. For example, the drain D.sub.24 of the support transistor **24** electrically couples to the back gate BG.sub.12 of the BioFET **12** and the source S.sub.24 of the support transistor **24** electrically couples to the controller **34** of the ASIC **30**.

[0162] The TIA **33** includes an operational amplifier **331** and a resistor **332**. The operational amplifier **331** includes a positive input terminal, a negative input terminal, an output terminal, a supply terminal, and a reference terminal. The supply terminal is configured to receive a supply voltage V.sub.DD, and the reference terminal is configured to receive the second voltage V**2**. As noted above, the second voltage V**2** may, for example, be ground. The positive input terminal is electrically coupled (e.g., shorted) to the reference terminal. The negative input terminal is electrically coupled to an output of the pixel circuit (e.g., a source S.sub.25 of the select transistor **25**). The output terminal is electrically coupled to an input of the ADC **31**, and the resistor **332** is electrically coupled from the negative input terminal to the output terminal. In alternative embodiments, the TIA **33** may have some other suitable structure.

[0163] In some embodiments, during use of the biosensor, a sensing well of the BioFET **12** is exposed to an aqueous fluid that may contain DNA segments, as well as auxiliary components such as primers, polymerase, and deoxynucleoside triphosphates (dNTPs). A reference bias is applied to the reference electrode **13**. If DNA segments are present in the fluid, thermal cycling is performed to replicate the DNA segments and increase their concentration for more reliable detection. The binding of DNA segments to a bio-sensing film at a channel of the BioFET **12** releases hydrogen ions into the fluid, increasing the pH in the fluid and altering the conductance of the channel. Therefore, the change in conductance of the channel is proportional to the concentration of DNA segments in the fluid. While this example demonstrates detection through pH variations, it is appreciated that different biological entities and detection mechanisms are amenable and can be used for various applications.

[0164] With reference to FIGS. **13**A and **13**B, circuit diagrams **1300**A, **1300**B of some alternative embodiments of the biosensor of FIG. **12** are provided. In FIG. **13**A, the support transistor **24** is instead electrically coupled to the channel C.sub.12 of the BioFET **12**. In FIG. **13**B, the support transistor **24** is instead electrically coupled to the reference electrode **13**.

[0165] The support transistor **24** is configured to selectively electrically couple the channel C.sub.12 or the reference electrode **13** to the fourth voltage V**4** between sensing cycles to repel charge away from the channel C.sub.12 of the BioFET **12**. As noted above, such ions and/or charges influence the conductance of the channel C.sub.12 and hence skew measurements by the BioFET **12**. Further, rather than receiving the fourth voltage V**4** via the controller **34**, the support transistor **24** may receive the fourth voltage V**4** directly. In alternative embodiments, the fourth voltage V**4** is still received via the controller **34**, similar to FIG. **12**.

[0166] With reference to FIG. **13**C, a circuit diagram **1300**C of some alternative embodiments of the biosensor of FIG. **12** is provided in which the controller **34** of the ASIC **30** is instead in the second IC die **2**. As a result, the ASIC **30** spans the second and third IC dies **2**, **3**.

[0167] With reference to FIG. **13**D, a circuit diagram **1300**D of some alternative embodiments of the biosensor of FIG. **12** is provided in which the BioFET **12** is configured to receive the second voltage V**2**, instead of the first voltage V**1**, at the second terminal (e.g., source S**12**) of the BioFET **12**. The second terminal may, for example, be electrically coupled (e.g., shorted) to individual transmit bodies of the pixel circuit **20** and the ASIC **30**.

[0168] With reference to FIG. **13**E, a circuit diagram **1300**E of some alternative embodiments of the biosensor of FIG. **12** is provided in which the second and third IC dies **2**, **3** are bonded and electrically coupled together by the second interconnect structure **42** instead of by the internal wire bond structure **43**. Hence, the first, second, and third IC dies **1**, **2**, **3** may be stacked together in a common 3D IC die or some other suitable structure.

[0169] With reference to FIG. **13**F, a circuit diagram **1300**F of some alternative embodiments of the biosensor of FIG. **12** is provided in which the TIA **33** is further configured to receive the first voltage V**1**, instead of the second voltage V**2**, at its reference terminal. As noted above, the first voltage V**1** may, for example, be less (e.g., negative) than the second voltage V**2**. Hence, the voltage across the operational amplifier **331** may be larger and the output range of the TIA may be larger than it would otherwise be. This may, for example, enhance performance.

[0170] With reference to FIG. **13**G, a circuit diagram **1300**G of some alternative embodiments of the biosensor of FIG. **13**F is provided in which the TIA **33** is in the first IC die **1**. Hence, the ASIC **30** is split amongst the first and third IC dies **1**, **3**.

[0171] With reference to FIG. **14**, a circuit diagram **1400** of some embodiments of the image sensors of FIG. **1** is provided in which the ADC **31** and the controller **34** comprise individual transistors. The ADC **31** comprises a plurality of transistors **311** (only one of which is shown) that are interconnected to implement functions of the ADC **31**. The plurality of transistors **311** comprise individual drains D**311**, individual sources S**311**, individual bodies B**311**, and individual gates G**311**. The individual bodies B**311** are configured to receive the second reference voltage V**2** and are on the substrate of the third IC die **3**. Hence, the individual bodies are biased by the second reference voltage V**2** during use of the biosensor.

[0172] Similar to the ADC **31**, the controller **34** comprises a plurality of transistors **341** (only one of which is shown) that are interconnected to implement functions of the controller **34**. The plurality of transistors **341** comprise individual drains D.sub.341, individual sources S.sub.341, individual bodies B.sub.341, and individual gates G.sub.341. The individual bodies B.sub.341 are configured to receive the second reference voltage V**2** and are on the substrate of the third IC die **3**. Hence, the individual bodies are biased by the second reference voltage V**2** during use of the biosensor.

[0173] While not shown, the operational amplifier **331** of the TIA **33** may also comprise individual transistors that are interconnected to implement functions of the operational amplifier **331**. These transistors may be similar to the transistors of the ADC **31** and the controller **34** and may hence have individual bodies configured to receive the second reference voltage V**2**.

[0174] With reference to FIG. **15**, a cross-sectional view **1500** of some embodiments of the biosensor of FIG. **12** is provided. The first and second IC dies **1**, **2** are stacked and bonded together on a package substrate **1502**. Such bonding may, for example, be via the first interconnect structure **41** of FIG. **12**, which may also provide electrical coupling. The third IC die **3** is on the package substrate **1502**, laterally adjacent to the first and second IC dies **1**, **2**. Further, the third IC die **3** is electrically coupled to the first and second IC dies **1**, **2** by the internal wire bond structure **43**, which extends from the third IC die **3** to the first IC die **1**.

[0175] The package substrate **1502** underlies and is bonded to the second and third **2**, **3** by adhesive layers **1504**. The package substrate **1502** may, for example, correspond to a printed circuit board (PCB) **1506** or some other suitable type of substrate. In alternative embodiments, the package substrate **1502** may, for example, be or comprise silicon and/or may, for example, correspond to a silicon interposer structure.

[0176] A plurality of package pads **1508** are on a top of the package substrate **1502** and a bottom of the package substrate **1502**. Further, a plurality of package vias **1510** extend through the package substrate **1502**, between package pads on a top of the package substrate **1502** and package pads on a bottom of the package substrate **1502**. The package pads on the package substrate **1502** may, for example, be configured to receive voltages for powering and/or controlling the biosensor. Such voltages may, for example, include any one or combination of the voltages V**1**, V**2**, V**4**, V**5**, and/or some other suitable voltages.

[0177] A plurality of external wire bond structures **44** extend from package pads on the top of the package substrate **1502** respectively to the third IC die **3** and the first IC die **1** to provide external connectivity to these dies. The plurality of external wire bond structures **44** may, for example, interface with package pads and/or the IC dies via bump structures **1516**. The bump structures **1516** may, for example, be solder bumps or some other suitable type of structure.

[0178] In some embodiments, the plurality of package vias **1510** surround corresponding dielectric cores **1512** extending through the package substrate **1502**. Further, in some embodiments, the plurality of package vias **1510** are separated from the package substrate **1502** by a dielectric layer **1514**. The dielectric layer **1514** may, for example, be localized to the plurality of package vias **1510** or may surround the package substrate **1502** as shown.

[0179] A lid **1518** overlies the package substrate **1502** and the first, second, and third IC dies **1**, **2**, **3**. Further, the lid **1518** surrounds the first, second, and third IC dies **1**, **2**, **3**. The lid **1518** has one or more holes **1518***h*, which overlie the first IC die **1** and are aligned with fluidic channels **1520** formed by a channel structure **1522**. The one or more holes **1518***h* are configured to receive fluid with entities for detection by the BioFETs of the first IC die **1**, and the fluidic channels **1520** are configured to carry the fluid to the BioFETs.

[0180] With reference to FIG. **16**A, a cross-sectional view **1600**A of some embodiments of the first IC die **1** and the second IC die **2** in the biosensor of FIG. **15** is provided. The biosensor includes an array of pixels, each pixel being as in any one or combination of FIGS. **12**, **13**A-**13**G, and **14**. The first and second IC dies **1**, **2** are vertically stacked with. The first IC die **1** comprises a semiconductor substrate **1***s* accommodating BioFETs **12**, whereas the second IC die **2** comprises a semiconductor substrate **2***s* accommodating pixel circuits **20**.

[0181] The first interconnect structure **41** is between the first and second IC dies **1**, **2** and corresponds to a bond structure. The bond structure includes a plurality of first conductive bond layers **411** and a plurality of first conductive bond vias **412** in a dielectric structure **1***d* of the first IC die **1** and further includes a plurality of second conductive bond layers **413** and a plurality of second conductive bond vias **414** in a dielectric structure **2***d* of the second IC die **2**. The plurality of first conductive bond layers **411** are connected to the plurality of second conductive bond layers **413**. The plurality of first conductive bond layers **411** and the plurality of second conductive bond layers **413** may be bonded by metal-to-metal bonding, whereas surrounding dielectric layers **1***d* and **2***d* may be bonded by dielectric-to-dielectric bonding.

[0182] The first IC die **1** includes a plurality of conductive wire levels M**11**, M**12**, M**13**, M**14** and one or more conductive via levels V**11**, V**12**, V**13** in the dielectric structure **1***d* of the first IC die **1**. Although FIG. **16**A shows four conductive wire levels and three conductive via level, more conductive wire levels and/or more conductive via levels are amenable. The first IC die **1** further includes a plurality of conductive contacts C**11** in the dielectric structure **1***d*. The plurality of conductive contacts C**11** electrically couple conductive wire level M**11** to the BioFETs **12** and to a bulk of the semiconductor substrate **1**s of the first IC die **1**.

[0183] The BioFETs **12** are in the semiconductor substrate **1***s* of the first IC die **1** and further include channel regions C.sub.12 and source/drain regions S.sub.12, D.sub.12 in the semiconductor substrate **1***s*. The reference electrode **13** is shared amongst the BioFETs **12** and is formed over the semiconductor substrate **1***s*, separated from the semiconductor substrate **1***s* by a passivation layer **1602**. In alternative embodiments, the reference electrode **13** is external to the semiconductor structure of FIG. **16**A and/or is individual to the BioFETs **12**.

[0184] The passivation layer **1602** forms sensing wells **1604**, which are individual to and respectively overlie the BioFETs **12**. Further, the sensing wells **1604** are lined by a sensing layer **1606** configured to bind with molecules carried in a fluid. Such molecules include, for example, DNA, proteins, cells, and so. The binding triggers the release of ions into the fluid, which alters channel conductivity of the BioFETs **12** and provides measurable signals indicative of the presence and quantity of bound bioentities. The sensing wells **1604** underlie fluidic channels **1520** configured to carry the fluid and formed by the channel structure **1522**, which overlies the sensing layer **1606**. Further, in some embodiments, the sensing layer **1606** has holes **1606***h* exposing the reference electrode **13** to the fluidic channels **1520**.

[0185] The second IC die **2** includes a plurality of conductive wire levels M**21**, M**22**, M**23**, M**24** and one or more conductive via levels V**21**, V**22**, V**23** in the dielectric structure **2***d* of the second IC die **2**. Although FIG. **16**A shows four conductive wire levels and three conductive via level, the second IC die **2** may include more conductive wire levels and/or more conductive via levels. The second IC die **2** further includes a plurality of conductive contacts C**21** in the dielectric structure **2***d*. The conductive contacts C**21** connect conductive wire level M**21** to a bulk of the semiconductor substrate **2***s* of the second IC die **2** and to the pixel circuits **20**.

[0186] The pixel circuits **20** comprise a plurality of transistors **20***t*, which correspond to the support transistors **24** and the select transistors **25** previously discussed. The plurality of transistors **20***t* may, for example, each be connected to conductive wire level M**21** by the plurality of conductive contacts C**21**.

[0187] The BioFETs **12** and the pixel circuits **20** are interconnected by the wire and via levels (e.g., conductive wire level M**21**, via level V**11**, etc.) and the first interconnect structure **41** to form the pixel array. Each pixel PX includes a BioFET **12** and a pixel circuit **20** and may, for example, be as in any one or combination of FIGS. **12**, **13**A-**13**G, and **14**.

[0188] The first IC die **1** further includes bump structures **1516** extending through the passivation layer **1602** and the semiconductor substrate **1**s of the first IC die **1**. The bump structures **1516** electrically couple wire bond structures **43**, **44** to conductive wire level M**11**. Through these wire bond structures, the first and second IC dies **1**, **2** receive various voltages, signals, and so on. Note that because FIG. **16**A is a cross-sectional view **1600**A, there may be additional wire bond structures outside of cross-sectional view **1600**A.

[0189] With reference to FIGS. **16**B, a cross-sectional view **1600**B of some embodiments of the third IC die **3** in the biosensor of FIG. **15** is provided.

[0190] The third IC die **3** includes a plurality of conductive wire levels M**31**, M**32**, M**33** and a plurality of conductive via levels V**31**, V**32** in the dielectric structure **3***d* of the third IC die **3**. Although FIG. **16**B shows three conductive wire levels and two conductive via levels, the third IC die **3** may include more conductive wire levels and/or more conductive via levels. The third IC die **3** further includes a plurality of conductive contacts C**31**.

[0191] The plurality of conductive contacts C**31** are in the dielectric structure **3***d* of the third IC die **3** and connect conductive wire level M**31** to a bulk of the semiconductor substrate **3***s* of the third IC die **3** and to the ASIC **30**. The ASIC **30** comprises individual transistors **30***t*, which form the ADC **31**, the TIA **33**, and the controller **34** previously discussed and may each be connected to conductive wire level M**31** by the plurality of conductive contacts C**31**.

[0192] The third IC die **3** further includes bump structures **1516** extending through into the dielectric structure **3***d* of the third IC die **3** to conductive wire level M**33** (e.g., a topmost wire level). The bump structures **1516** electrically couple wire bond structures **43**, **44** to conductive wire level M**33**. Through these wire bond structures, the third IC die **1** receives various voltages, signals, and so on and outputs various voltages, signals and so on to the first and second IC dies **1**, **2**. Note that because FIG. **16**B is a cross-sectional view **1600**B, there may be additional wire bond structures outside of cross-sectional view **1600**B.

[0193] As previously discussed, the BioFETs **12** are configured to receive the first reference voltage V**1**. In FIG. **16**A, the biosensor is configured to receive the first reference voltage V**1** via a wire bond structure. Conductive features (e.g., wires, vias, contacts, etc.) of the first IC die **1** then pass the first reference voltage V**1** to the semiconductor substrate **1***s* of the first IC die **1** and/or to the BioFETs **12** (e.g., to sources S**12**).

[0194] Also, as previously discussed, the pixel circuit **20** and the ASIC **30** are configured to receive the second reference voltage V**2**. In FIGS. **16**A and **16**B, the pixel circuit **20** and the ASIC **30** are configured to receive the second reference voltage V**2** via corresponding wire bond structures. Conductive features (e.g., wires, vias, contacts, interconnect structures, etc.) in the first and second IC dies **1**, **2** pass the second reference voltage V**2** to the semiconductor substrate **2***s* of the second IC die **2**. The semiconductor substrate **2***s* of the second IC die **2** forms one or more conductive paths to pass the second reference voltage V**2** to the transistors **20***t* of the pixel circuit **20** (e.g., bodies of the transistors **20***t*).

[0195] Similar to the first and second IC die **1**,**2**, conductive features (e.g., wires, vias, contacts, interconnect structures, etc.) in the third IC die **3** pass the second reference voltage V**2** to the semiconductor substrate **3***s* of the third IC die **3**. The semiconductor substrate **3***s* of the third IC die **3** forms one or more conductive paths to pass the second reference voltage V**2** to the transistors **30***t* of the ASIC **30** (e.g., bodies of the transistors **30***t*).

[0196] With reference to FIG. **17**, a plan view **1700** of some embodiments of the first IC die **1** of FIG. **16**A is provided in which the reference electrode **13** and the BioFETs **12** are shown in phantom. The reference electrode **13** is shared by the BioFETs **12** and has an inverted U-shaped profile. Alternatively, the BioFETs **12** may have individual reference electrodes for the BioFETs **12** and/or the reference electrode **13** may have some other suitable shape. The BioFETs **12** are arranged in three rows and two columns. However, there may be more or less rows and more or less columns. Each BioFET **12** has its own sensing well **1604** and is proximate to its own hole **1606***h* in the sensing layer **1606**. As noted above, such holes expose the reference electrode **13**. A plurality of bump structures **1516** are arranged around the periphery of the first IC die to receive voltages, signals, and so on from outside the biosensor and/or from the ASIC **30**.

[0197] With reference to FIG. **18**, a cross-sectional view **1800** of some alternative embodiments of the biosensor of FIG. **15** is provided in which the third IC die **3** is stacked with and electrically coupled to the first and second IC dies **1**, **2**. Further, the second IC die **2** is between the first and third IC dies **1**, **3**. As should be appreciated, this stacked structure is similar to the stacked structure described for the image sensor of FIG. **4**.

[0198] With reference to FIG. **19**, a cross-sectional view **1900** of some embodiments of the first, second, and third IC dies **1-3** in the biosensor of FIG. **18** is provided. The first, second, and third IC dies **1-3** are similar to the first, second, and third IC dies **1-3** in FIGS. **16**A and **16**B, except that the third IC die **3** is now bonded to the second IC die **2**. Further, the third IC die **3** is electrically coupled to the second IC die **2** by a second interconnect structure **42**.

[0199] The second interconnect structure **42** corresponds to a plurality of conductive through vias. The plurality of conductive through vias extend through the semiconductor substrate **2***s* of the second IC die **2** and are separated from the semiconductor substrate **2***s* by dielectric liners **421**. The conductive through vias may be copper, gold, aluminum, or the like.

[0200] With reference to FIG. **20**, a cross-sectional view **2000** of some alternative embodiments of the biosensor of FIG. **15** is provided in which the second and third IC dies **2**, **3** are stacked together and laterally offset from the first IC die **1**. Further, the first IC die **1** is electrically coupled to the second and third IC dies **2**, **3** by an internal wire bond structure **45**.

[0201] With reference to FIG. **21**A, a cross-sectional view **2100**A of some embodiments of the first IC die **1** in the biosensor of FIG. **20** is provided. The first IC die **1** is similar to the first IC die **1** in FIG. **16**A, except that that first IC die **1** is now bonded to a carrier substrate **2102** rather than the second IC die **2**. The carrier substrate **2102** may, for example, be or comprise a silicon substrate or some other suitable type of substrate.

[0202] With reference to FIG. **21**B, a cross-sectional view **2100**B of some embodiments of the second and third IC dies **2**, **3** in the biosensor of FIG. **20** is provided. The second and third IC dies **2**, **3** are similar to the second and third IC dies **2**, **3** in FIGS. **16**A and **16**B, respectively. However, the second IC die **2** is vertically flipped and bonded to and electrically coupled to the third IC dies **2**, **3** via a third interconnect structure **46**.

[0203] The third interconnect structure corresponds to a bond structure. The bond structure comprises a plurality of first conductive bond layers **461** and a plurality of first conductive bond vias **462** in a dielectric structure **2***d* of the second IC die **2** and further includes a plurality of second conductive bond layers **463** and a plurality of second conductive bond vias **464** in a dielectric structure **3***d* of the third IC die **3**.
[0204] The plurality of first conductive bond layers **461** are connected to the plurality of second conductive bond layers **463**. The plurality of first conductive bond layers **461** and the plurality of second conductive bond layers **463** may be bonded by metal-to-metal bonding, whereas surrounding dielectric layers **2***d* and **3***d* may be bonded by dielectric-to-dielectric bonding. Conductive wire level M**23** is connected to the first conductive bond vias **462**, and conductive wire level M**33** is connected to the second conductive bond vias **464**.
[0205] A plurality of conductive pads **2104** are inset into the semiconductor substrate **2***s* of the second IC die **2** and have individual legs or protrusions extending to conductive wire level M**23**. The individual legs or protrusions extend through a trench isolation structure **2106** and a passivation layer **2108**, the latter of which separates the plurality of conductive pads **2104** from the semiconductor substrate **2***s* of the second IC die **2**.
[0206] A plurality of bump structures **1516** are on the plurality of conductive pads **2104** to electrically couple the plurality of conductive pads to wire bond structures **44**, **45**. Via the bump structures **1516**, the plurality of conductive pads **2104**, and the wire bond structures **44**, **45**, the second and third IC dies **2**, **3** may receive voltages, signals, and so and an may pass voltages, signals, and so to the first IC die **1**. The bump structures **1516** may, for example, be solder bumps and/other suitable types of bump structures.
[0207] With reference to FIG. **22**, a circuit diagram **2200** of some embodiments of a biosensor that spans two IC dies, according to aspects of the present disclosure, is provided. The biosensor is similar to the biosensor of FIG. **12**. However, the pixel circuit **20** and the ASIC **30** are both in the second IC die **2** and the third IC die **3** is omitted. Because the pixel circuit **20** and the ASIC **30** have the same reference voltage (e.g., voltage V**2**), variations in reference voltage are the same in the pixel circuit **20** and the ASIC **30**. As a result, noise is reduced and the SNR of the ADC **31** is enhanced. Further, because the BioFET **12** has the first reference voltage V**1**, which is less than the second reference voltage V**2**, the voltage across the BioFET **12** may be larger than it would otherwise be and sensitivity may be enhanced.
[0208] With reference to FIGS. **23**A-**23**E, various circuit diagrams **2300**A-**2300**E of some alternative embodiments of the biosensor of FIG. **22** are provided. FIGS. **23**A and **23**B are similar respectively to FIGS. **13**A and **13**B, FIG. **23**C is similar to FIG. **13**D, and FIGS. **23**D and **23**E are similar respectively to FIGS. **13**F and **13**G. However, the pixel circuit **20** and the ASIC **30** are in the second IC die **2** and the third IC die **3** is omitted.
[0209] In FIG. **23**A, the support transistor **24** is electrically coupled to the back gate BG.sub.12 of the BioFET **12** to repel charge away from the channel C.sub.12 of the BioFET **12** between sensing cycles. In FIG. **23**B, the support transistor **24** is electrically coupled to the reference electrode **13** to repel charge away from the channel C.sub.12 of the BioFET **12** between sensing cycles.
[0210] In FIG. **23**C, the BioFET **12** has the same reference voltage as the pixel circuit **20** and the ASIC **30**. For example, the second terminal (e.g., the source S**12**) and individual transistor bodies of the pixel circuit **20** and the ASIC **30** may be configured to receive the second reference voltage V**2** and may hence be electrically coupled (e.g., shorted together).
[0211] In FIGS. **23**D and **23**E, the TIA **33** is further configured to receive the first voltage V**1**, instead of the second voltage V**2**, at its reference terminal. Further, the TIA **33** is in the second IC die **2** in FIG. **23**D and is in the first IC die **1** in FIG. **23**E.
[0212] With reference to FIG. **24**, a circuit diagram **2400** of some embodiments of the biosensor of FIG. **22** is provided in which the controller **34** and the ADC **31** further comprise individual transistors. The ADC **31** comprises a plurality of transistors **311** (only one of which is shown) that

are interconnected to implement functions of the ADC **31**. Similar to the ADC **31**, the controller **34** comprises a plurality of transistors **341** (only one of which is shown) that are interconnected to implement functions of the controller **34**. Further, individual bodies of the transistors **311** of the ADC **31** and the transistors **341** of the controller are configured to receive the second reference voltage V**2** and are electrically coupled (e.g., shorted) together.

[0213] While not shown, the operational amplifier **331** of the TIA **33** may also comprise individual transistors that are interconnected to implement functions of the operational amplifier **331**. These transistors may be similar to the transistors of the ADC **31** and the controller **34** and may hence have individual bodies configured to receive the second reference voltage V**2**.

[0214] With reference to FIG. **25**, a cross-sectional view **2500** of some embodiments of the biosensor of FIG. **22** is provided. The biosensor is similar to FIG. **15**, except that the third IC chip **3** is omitted. Further, the ASIC **30** (not shown) is in the second IC die **2**.

[0215] With reference to FIG. **26**, a cross-sectional view **2600** of some embodiments a first IC die **1** and a second IC die **2** in the biosensor of FIG. **25** is provided. The first and second IC dies **1**, **2** similar to the first and second IC dies **1**, **2** in FIG. **16**A, except that that pixel circuits **20** and the ASIC **30** are both in the second IC die **2**.

[0216] While FIGS. **25** and **26** illustrate the first and second IC dies **1**, **2** as being stacked, the first and second IC dies **1**, **2** may alternatively be laterally spaced from each other. For example, similar to how the third IC die **3** is laterally spaced from the first and second IC dies **1**, **2** in FIG. **15**, the first and second IC dies **1**, **2** may be laterally spaced.

[0217] With reference to FIG. **27**A, a block diagram **2700**A of some embodiments of a biosensor comprising a plurality of pixels PX, each as in any one or combination of FIGS. **12**, **13**A-**13**G, **14**, **15**, **16**A, **16**B, **17-20**, **21**A, **21**B, **22**, **23**A-**23**E, and **24-25**, is provided. The plurality of pixels PX are in a plurality of rows and a plurality or columns to form a pixel array **2702**. For example, three rows and two columns are illustrated. However, there may be more or less rows and/or more or less columns in alternative embodiments.

[0218] The plurality of pixels PX comprise individual BioFETs **12** and individual pixel circuits **20**. Further, the plurality of pixels PX are shown as having individual reference electrodes **13** but may share a common reference electrode **13** in alternative embodiments. Further yet, the plurality of pixels PX share the ASIC **30** and are each electrically coupled to the ASIC **30**. Therefore, there may be only one ASIC **30** in at least some embodiments. As above, the BioFETs **12** are in the first IC die **1** with the first reference voltage V**1**, and the pixel circuits **20** are in the second IC die **2** with the second reference voltage V**2**. Further, the ASIC **30** is at least partially in a second or third IC die **2**, **3** with the second reference voltage V**2**.

[0219] With reference to FIG. **27**B, a circuit diagram **2700**B of the pixel array **2702** of FIG. **27**A is provided. The pixels PX are individually connected to a conductive line **2704** (e.g., via sources of select transistors **25** in the individual pixels PX). The conductive line **2704** extends to the ASIC **30** and electrically couples with the TIA **33** and the ADC **31**.

[0220] With reference to FIG. **28**, a block diagram **2800** of some embodiments of a method of use according to aspects of the present disclosure. The method of use is common to both the images sensors (e.g., the image sensor of FIG. **1**) and the biosensors (e.g., the biosensor of FIG. **12**). Further, the method is applicable to other pixel-based devices comprising a sensor and a pixel circuit electrically coupled to the sensor.

[0221] At Act **2802**, a semiconductor structure comprising a pixel is provided. The semiconductor structure comprises a sensor in a first IC die and a pixel circuit. The pixel circuit comprises a plurality of pixel transistors in a second IC die. The sensor comprises a first terminal and a second terminal both in a semiconductor substrate of the first IC die, and the first terminal is electrically coupled to an input of the pixel circuit.

[0222] At Act **2804**, a first voltage is applied to the second terminal of the sensor.

[0223] At Act **2806**, while applying the first voltage, a second voltage greater than the first voltage is applied to individual transistor bodies of the plurality of pixel transistors.

[0224] At Act **2808**, a signal based on a value sensed by the sensor is generated at an output of the pixel circuit.

[0225] In some embodiments, the sensor is a pinned photodetector. Such embodiments may, for example, arise when the method of use is applied to the image sensor in any one of FIGS. **1** to **11**B. In other embodiments, the sensor is a BioFET. Such other embodiments may, for example, arise when the method of use is applied to the biosensor in any one of FIGS. **12** to **27**B.

[0226] In some embodiments in which the sensor is a BioFET, the method of use further includes applying a voltage to a reference electrode of the BioFET, a channel of the BioFET, or a back gate of the BioFET between sensing cycles. A sensing cycle may be regarded as acts **2802** to **2808**. The voltage repels ions and/or charges near the channel that may linger from previous sensing cycles and skew measurements with the BioFET. For negative ions or charges, the voltage may be negative. For positive ions or charges, the voltage may be positive.

[0227] In some embodiments in which the sensor is a BioFET, the method of use further includes applying a voltage to the back gate of the BioFET during a sensing cycle. As above, a sensing cycle may be regarded as acts **2802** to **2808**. The voltage attracts ions and/or charges to channel to enhance sensitivity of the BioFET. For negative ions or charges, the voltage may be positive. For positive ions or charges, the voltage may be negative.

[0228] While the block diagram **2800** of FIG. **28** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

[0229] While the present disclosure focused on image sensors and biosensors, it is to be appreciated that the aforementioned arrangements and solutions are applicable to other device types. For example, circuit components can be shifted amongst different IC dies, and/or additional voltages can be added to other device types, to reduce noise and/or enhance performance. Such other device types include, for example, Microelectromechanical Systems (MEMS) microphones, three-dimensional (3D) memory, 3D fabric, photonics, and so on.

[0230] This approach of biasing stacked substrates with different voltages is applicable to 3D memory structures (e.g., high bandwidth memory, HBM, and so on). These structures may include vertically stacked memory dies (e.g., DRAM or the like) and a controller die, with independent supply voltages applied to each. Independent ground levels can be maintained, and periphery circuitry (e.g., sense amplifiers, data buffers, decoders, and so on) can be manufactured on a separate substrate and biased with the same voltage as the controller die. This enables cancellation of ground-level variations, reducing noise and improving SNR.

[0231] Similarly, this approach can be applied to 3D MEMS structures comprising functional dies (e.g., sensors, logic controllers, thermal dissipaters, power controllers, and so on) with independently biased substrates. Maintaining independent ground levels allows for noise reduction. Input/output (I/O) devices of a sensor die can be separately manufactured on another substrate and biased with the same voltage as the controller die's I/O devices, further enabling ground-level variation cancellation and SNR improvement.

[0232] Reducing the size of functional dies (e.g., X-Y area) is achievable by manufacturing I/O devices on a separate substrate or integrating them into the controller die. This also increases the utility area of the functional die by removing the I/O devices from its footprint.

[0233] In some embodiments, a MEMS structure is designed for sound detection. This structure comprises a substrate, a semiconductor die for control and signal processing, a sensor incorporating a transducer, and a protective lid. Sound waves enter a cavity defined by the substrate and lid,

causing a membrane within the sensor to vibrate. This vibration is converted into an electrical signal by a piezoelectric component and transmitted to the semiconductor die.
[0234] The substrate incorporates multiple conductive layers, vias, and pads to facilitate electrical connections and voltage delivery. The semiconductor die operates with a first bias voltage, establishing its grounding level. This voltage can also be used to power or ground the sensor. The lid is grounded through the substrate's conductive layers.
[0235] A challenge in transducer design is the accumulation of electrostatic charges on the vibrating membrane, leading to stiction and noise. To address this, the disclosed structure incorporates additional wiring and a second bias voltage applied to the transducer. This second voltage, differing from first bias voltage, releases the accumulated charges, minimizing stiction and improving signal clarity. The second bias voltage can be applied continuously or intermittently, depending on the application of the MEMS structure.
[0236] By carefully managing voltage levels and addressing the issue of electrostatic charges, this MEMS structure provides a robust and accurate solution for sound detection and other sensing applications. The principles outlined in this disclosure are applicable to a wide range of MEMS devices and can be adapted to enhance performance and minimize noise.
[0237] In some embodiments, a 3D memory structure, such as, for example, a HBM, comprises vertically stacked memory dies and a controller die. Independent supply voltages are applied to the substrates of both the memory dies and the controller die. Periphery circuitry, such as sense amplifiers decoders, and so on, is manufactured on a separate substrate or integrated directly into the controller die. This circuitry is biased with the same voltage as the controller die, effectively canceling out ground level variations and reducing noise, thereby improving the SNR of the controller.
[0238] In some embodiments, the memory dies and controller die are fabricated using different technology nodes or by different manufacturers and are biased with differing grounding levels. Dedicated transmission paths ensure appropriate voltage delivery to each component. The memory dies may, for example, utilize a negative grounding voltage, while the controller's logic devices may, for example, operate with a zero-volt ground. By relocating the periphery circuitry, the size of the memory dies can be reduced, increasing the usable area and overall efficiency of the 3D memory structure.
[0239] In some embodiments, a 3D fabric, such as Chip-on-Wafer-on-Substrate (CoWoS) or Integrated Fan-Out (InFO), utilizes independent grounding voltages for different dies—including memory, logic, and sensor dies—to enhance performance and address variations arising from differing manufacturing processes or suppliers.
[0240] Memory dies may, for example, be supplied with a negative grounding voltage to enhance performance, while logic and I/O dies may, for example, operate with a zero-volt ground. Sensor dies, such as those with analog components, may also utilize a negative grounding voltage to improve dynamic range. Multiple dies are interconnected via an interposer or redistribution layer, with dedicated transmission paths ensuring each die receives its designated grounding voltage. This approach mitigates ground tracking issues and allows for customized voltage levels tailored to each die's specific requirements. Further, by independently controlling the grounding voltages, the 3D fabric can enhance performance, reduce noise, and accommodate dies manufactured using different technology nodes or by different suppliers.
[0241] In some embodiments, a photonic device utilizes independent biasing of different substrates to enhance performance. This approach is applicable to optical communication systems and other photonic devices. The system comprises an optical source, a modulator, a receiver, and associated drivers and processors distributed across two substrates. The first substrate houses the optical components, while the second substrate contains the control electronics. The modulator includes multiple modulators (e.g., ring modulators or the like) with independently tunable resonant wavelengths. These wavelengths are adjusted using a combination of electrical and thermal drivers.

[0242] Independent grounding voltages are applied to the two substrates, allowing for customized biasing of the optical and electronic components. This enables enhancement of performance metrics such as dynamic range and free spectral range. In some embodiments, the modulator incorporates PIN phase shifters and metal heaters to control the optical signal. Independent biasing of these components allows for fine-tuning of the modulator's characteristics. By strategically applying different bias voltages to the optical and electronic components, the photonic device can achieve improved performance, reduced noise, and enhanced signal integrity.

[0243] In some embodiments, the present disclosure relates to a semiconductor structure, including: a first IC die including a first semiconductor substrate; a second IC die including a second semiconductor substrate; and a pixel including a sensor in the first IC die and a pixel circuit, wherein the pixel circuit includes a plurality of pixel transistors in the second IC die, the sensor includes a first terminal and a second terminal both in the first semiconductor substrate, the first terminal is electrically coupled to an input of the pixel circuit, and the second terminal is electrically isolated from individual bodies of the plurality of pixel transistors. In some embodiments, the sensor includes a photodetector in the first semiconductor substrate, wherein the pixel circuit further includes a transfer transistor in the first IC die and separating the sensor from a remainder of the pixel circuit. In some embodiments, the semiconductor structure further includes: a third IC die including an ADC, which has an input electrically coupled to an output of the pixel circuit; and a current source having a first terminal electrically coupled to the output of the pixel circuit and the input of the ADC and further having a second terminal electrically isolated from the second terminal of the sensor. In some embodiments, the current source is in the first IC die. In some embodiments, the sensor includes BioFET. In some embodiments, the semiconductor structure further includes: a third IC die including an ADC; and a TIA electrically coupled from an output of the pixel circuit to an input of the ADC, wherein the TIA is in the first IC die. In some embodiments, the semiconductor structure further includes: a third IC die including an ADC and a TIA electrically coupled from an output of the pixel circuit to an input of the ADC. The semiconductor structure further includes: a third IC die including a controller that is configured to selectively pass one of multiple different voltages to a channel of the BioFET or a back gate of the BioFET.

[0244] In other embodiments, the present disclosure relates to another semiconductor structure, including: a photodetector on a first semiconductor substrate; a pixel circuit including a transfer transistor on the first semiconductor substrate and a plurality of pixel transistors on a second semiconductor substrate, wherein the transfer transistor is electrically coupled to a cathode of the photodetector and separates the photodetector from the plurality of pixel transistors; and an ADC that is on the second semiconductor substrate and that is electrically coupled to an output of the pixel circuit, wherein the plurality of pixel transistors and a transistor of the ADC have individual transistor bodies electrically coupled together and isolated from an anode of the photodetector. In some embodiments, the semiconductor further includes a current source on the first semiconductor substrate and electrically coupled from the output of the pixel circuit to the anode of the photodetector. In some embodiments, the semiconductor further includes a current source on the second semiconductor substrate and electrically coupled from the output of the pixel circuit to the anode of the photodetector. In some embodiments, the semiconductor further includes a current source on the second semiconductor substrate, wherein the current source has a first terminal electrically coupled to the output of the pixel circuit and a second terminal electrically isolated from the individual transistor bodies. In some embodiments, the semiconductor further includes a current source on the second semiconductor substrate and including an additional transistor and a capacitor, wherein the additional transistor has a drain electrically coupled to the output of the pixel circuit, wherein the capacitor is electrically coupled from a gate of the additional transistor to a source of the additional transistor, and wherein the source of the additional transistor and a body of the additional transistor are electrically coupled together and electrically isolated from the individual transistor bodies. In some embodiments, the plurality of pixel transistors includes a source-follower transistor having a gate electrically coupled to a drain of the transfer transistor and further includes a select transistor electrically coupled from the source-follower transistor to the output of the pixel circuit.

[0245] In yet other embodiments, the present disclosure relates to a method, including: providing a semiconductor structure including a pixel, which includes a sensor in a first IC die and a pixel circuit, wherein the pixel circuit includes a plurality of pixel transistors in a second IC die, wherein the sensor includes a first terminal and a second terminal both in a semiconductor substrate of the first IC die, and wherein the first terminal is electrically coupled to an input of the pixel circuit; applying a first voltage to the second terminal of the sensor; while applying the first voltage, applying a second voltage greater than the first voltage to individual transistor bodies of the plurality of pixel transistors; and generating a signal based on a value sensed by the sensor at an output of the pixel circuit. In some embodiments, the sensor includes a pinned photodiode, wherein the second terminal of the sensor corresponds to a cathode of the pinned photodiode. In some embodiments, the sensor includes a BioFET, wherein the second terminal of the sensor corresponds to a source or drain region of the BioFET. In some embodiments, the method further includes, before generating the signal, applying a voltage to a back gate of the BioFET or a channel of the BioFET to clear charge in a sensing well of the BioFET from proximate the channel of the BioFET. In some embodiments, the method further includes, while generating the signal, applying a voltage to a back gate of the BioFET to attract charge in a sensing well of the BioFET to proximate a channel of the BioFET. In some embodiments, the first voltage is negative and the second voltage is ground.

[0246] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a first integrated circuit (IC) die comprising a first semiconductor substrate; a second IC die comprising a second semiconductor substrate; and a pixel comprising a sensor in the first IC die and a pixel circuit, wherein the pixel circuit comprises a plurality of pixel transistors in the second IC die, the sensor comprises a first terminal and a second terminal both in the first semiconductor substrate, the first terminal is electrically coupled to an input of the pixel circuit, and the second terminal is electrically isolated from individual bodies of the plurality of pixel transistors.

2. The semiconductor structure according to claim 1, wherein the sensor comprises a photodetector in the first semiconductor substrate, and wherein the pixel circuit further comprises a transfer transistor in the first IC die and separating the sensor from a remainder of the pixel circuit.

3. The semiconductor structure according to claim 1, further comprising: a third IC die comprising an analog-to-digital converter (ADC), which has an input electrically coupled to an output of the pixel circuit; and a current source having a first terminal electrically coupled to the output of the pixel circuit and the input of the ADC and further having a second terminal electrically isolated from the second terminal of the sensor.

4. The semiconductor structure according to claim 3, wherein the current source is in the first IC die.

5. The semiconductor structure according to claim 1, wherein the sensor comprises a bio field-effect transistor (BioFET).
6. The semiconductor structure according to claim 5, further comprising: a third IC die comprising an analog-to-digital converter (ADC); and a transimpedance amplifier (TIA) electrically coupled from an output of the pixel circuit to an input of the ADC, wherein the TIA is in the first IC die.
7. The semiconductor structure according to claim 5, further comprising: a third IC die comprising an analog-to-digital converter (ADC) and a transimpedance amplifier (TIA) electrically coupled from an output of the pixel circuit to an input of the ADC.
8. The semiconductor structure according to claim 5, further comprising: a third IC die comprising a controller that is configured to selectively pass one of multiple different voltages to a channel of the BioFET or a back gate of the BioFET.
9. A semiconductor structure, comprising: a photodetector on a first semiconductor substrate; a pixel circuit comprising a transfer transistor on the first semiconductor substrate and a plurality of pixel transistors on a second semiconductor substrate, wherein the transfer transistor is electrically coupled to a cathode of the photodetector and separates the photodetector from the plurality of pixel transistors; and an analog-to-digital converter (ADC) that is on the second semiconductor substrate and that is electrically coupled to an output of the pixel circuit, wherein the plurality of pixel transistors and a transistor of the ADC have individual transistor bodies electrically coupled together and isolated from an anode of the photodetector.
10. The semiconductor structure according to claim 9, further comprising: a current source on the first semiconductor substrate and electrically coupled from the output of the pixel circuit to the anode of the photodetector.
11. The semiconductor structure according to claim 9, further comprising: a current source on the second semiconductor substrate and electrically coupled from the output of the pixel circuit to the anode of the photodetector.
12. The semiconductor structure according to claim 9, further comprising: a current source on the second semiconductor substrate, wherein the current source has a first terminal electrically coupled to the output of the pixel circuit and a second terminal electrically isolated from the individual transistor bodies.
13. The semiconductor structure according to claim 9, further comprising: a current source on the second semiconductor substrate and comprising an additional transistor and a capacitor, wherein the additional transistor has a drain electrically coupled to the output of the pixel circuit, wherein the capacitor is electrically coupled from a gate of the additional transistor to a source of the additional transistor, and wherein the source of the additional transistor and a body of the additional transistor are electrically coupled together and electrically isolated from the individual transistor bodies.
14. The semiconductor structure according to claim 9, wherein the plurality of pixel transistors comprises a source-follower transistor having a gate electrically coupled to a drain of the transfer transistor and further comprises a select transistor electrically coupled from the source-follower transistor to the output of the pixel circuit.
15. A method, comprising: providing a semiconductor structure comprising a pixel, which comprises a sensor in a first integrated circuit (IC) die and a pixel circuit, wherein the pixel circuit comprises a plurality of pixel transistors in a second IC die, wherein the sensor comprises a first terminal and a second terminal both in a semiconductor substrate of the first IC die, and wherein the first terminal is electrically coupled to an input of the pixel circuit; applying a first voltage to the second terminal of the sensor; while applying the first voltage, applying a second voltage greater than the first voltage to individual transistor bodies of the plurality of pixel transistors; and generating a signal based on a value sensed by the sensor at an output of the pixel circuit.
16. The method according to claim 15, wherein the sensor comprises a pinned photodiode, and wherein the second terminal of the sensor corresponds to a cathode of the pinned photodiode.

17. The method according to claim 15, wherein the sensor comprises a bio field-effect transistor (BioFET), and wherein the second terminal of the sensor corresponds to a source or drain region of the BioFET.

18. The method according to claim 17, further comprising: before generating the signal, applying a voltage to a back gate of the BioFET or a channel of the BioFET to clear charge in a sensing well of the BioFET from proximate the channel of the BioFET.

19. The method according to claim 17, further comprising: while generating the signal, applying a voltage to a back gate of the BioFET to attract charge in a sensing well of the BioFET to proximate a channel of the BioFET.

20. The method according to claim 15, wherein the first voltage is negative and the second voltage is ground.